上海科技大学
ShanghaiTech University

CS177 Bioinformatics: Software Development and Applications

Lecture 21: AI for Drug Design (AIDD)

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Spring, 2025

Credit: PPT slides courtesy from Prof. Fang Bai, SIAIS, SLST, SIST at ShanghaiTech

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Outline

- Overview of drug discovery pipeline
- AI for drug development
- Drug target
- Drug-target interaction (DTI)
- Drug action mechanisms
- Protein structure
- ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity)

Overview of Drug Discovery Pipeline

Driving force for drug development

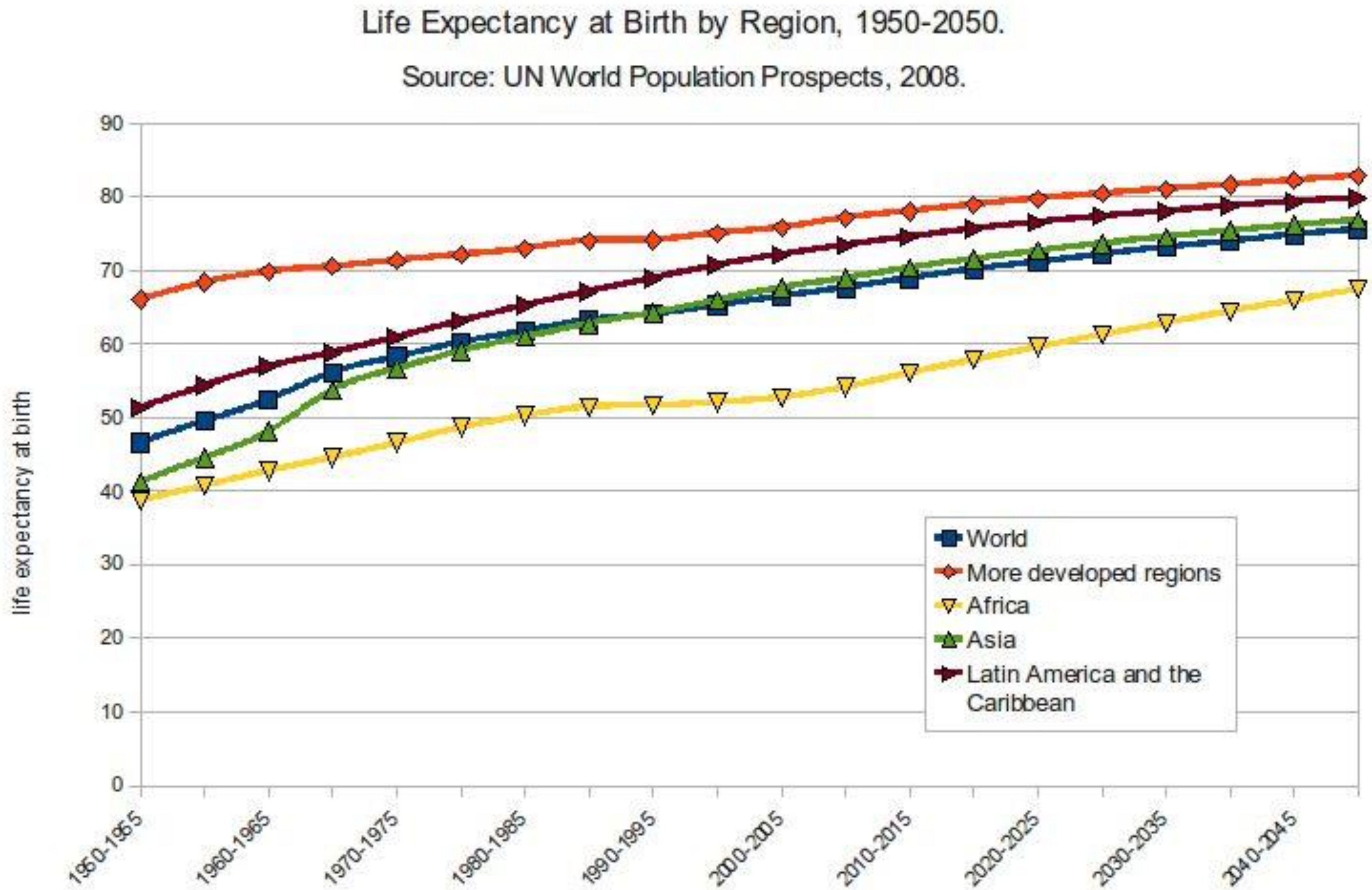
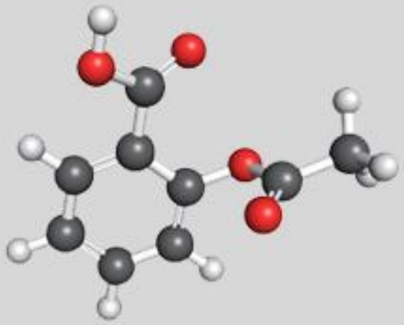
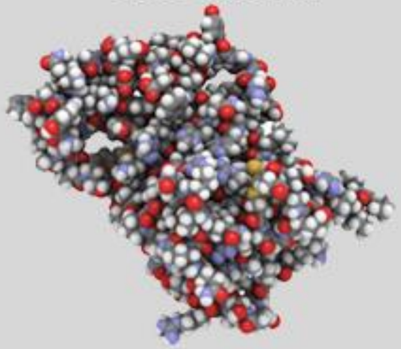
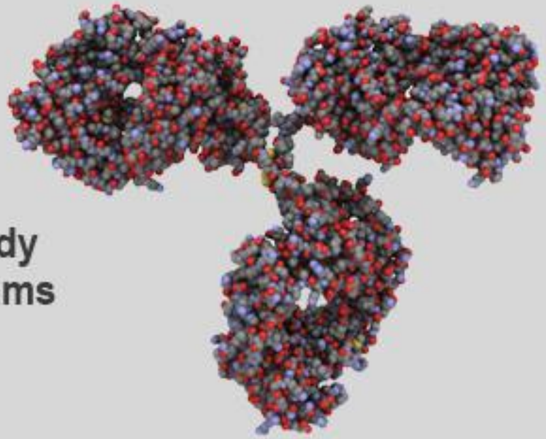


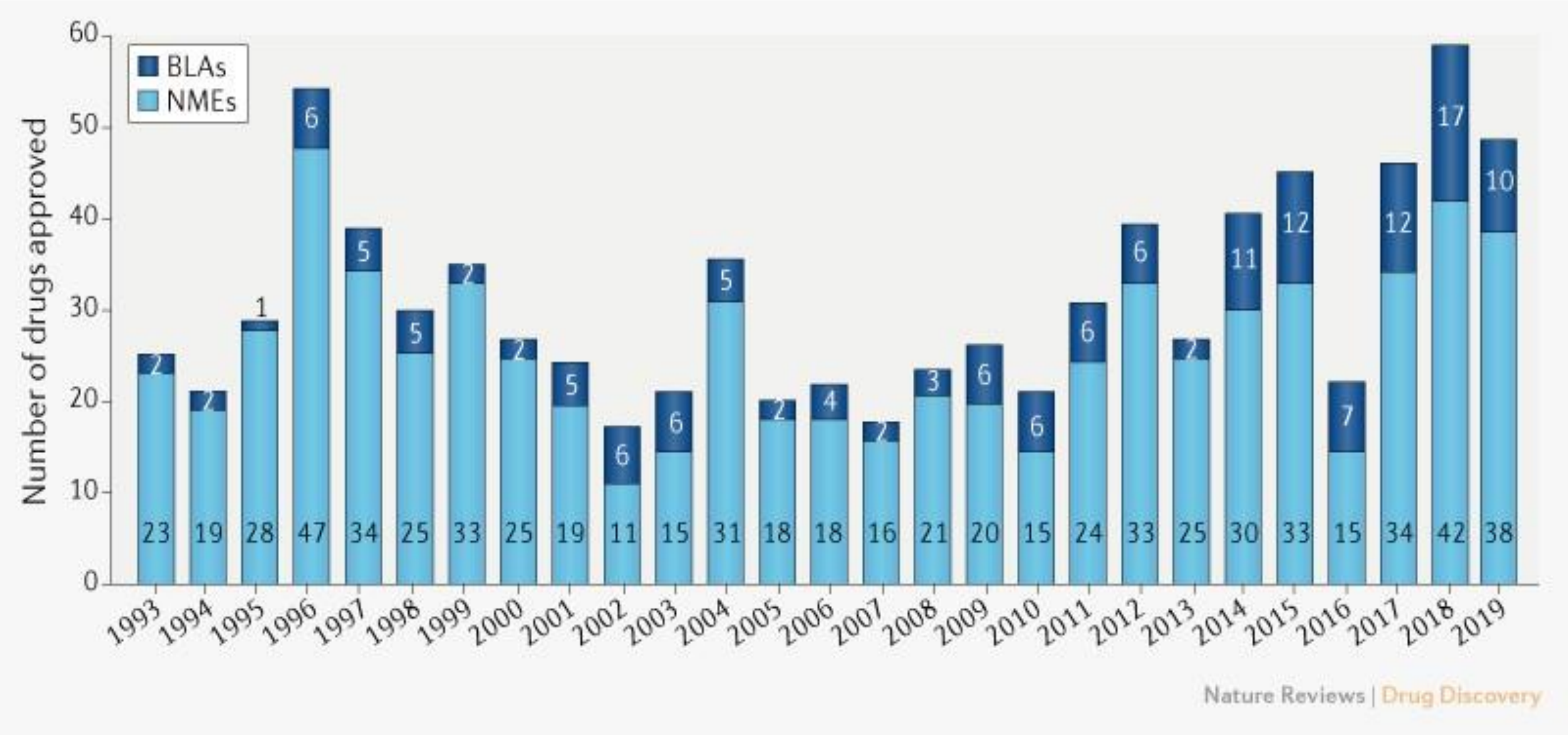
Photo by Rcragun CC-BY 3.0

Size & complexity of small molecular drugs / biological medicine

	Small Molecule Drug	Large Molecule Drug	Large Biologic
Size	Aspirin 21 atoms 	hGH ~ 3000 atoms 	 IgG Antibody ~ 25,000 atoms
Complexity	Bike ~ 20 lbs 	Car ~ 3000 lbs 	 Business Jet ~ 30,000 lbs (without fuel)

阿司匹林　人体生长激素　免疫球蛋白

Novel FDA approvals since 1993

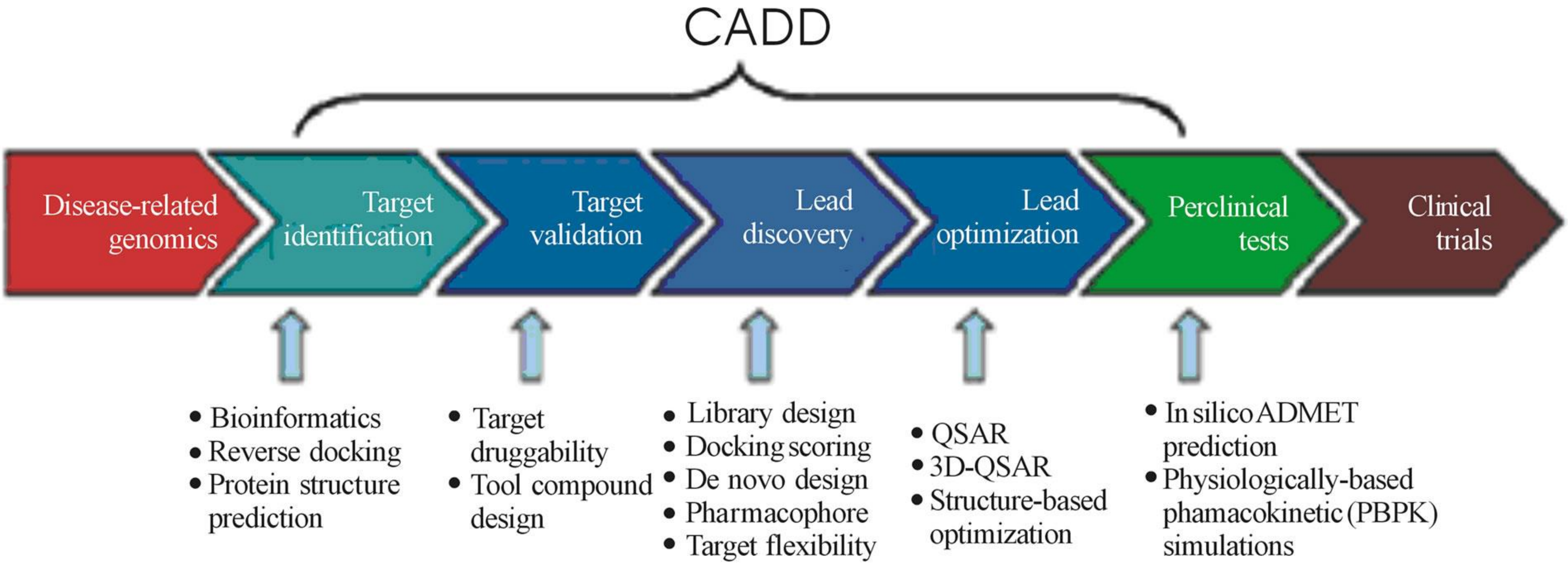


Annual numbers of new molecular entities (NMEs) and biologics license applications (BLAs) approved by the FDA's Center for Drug Evaluation and Research (CDER).

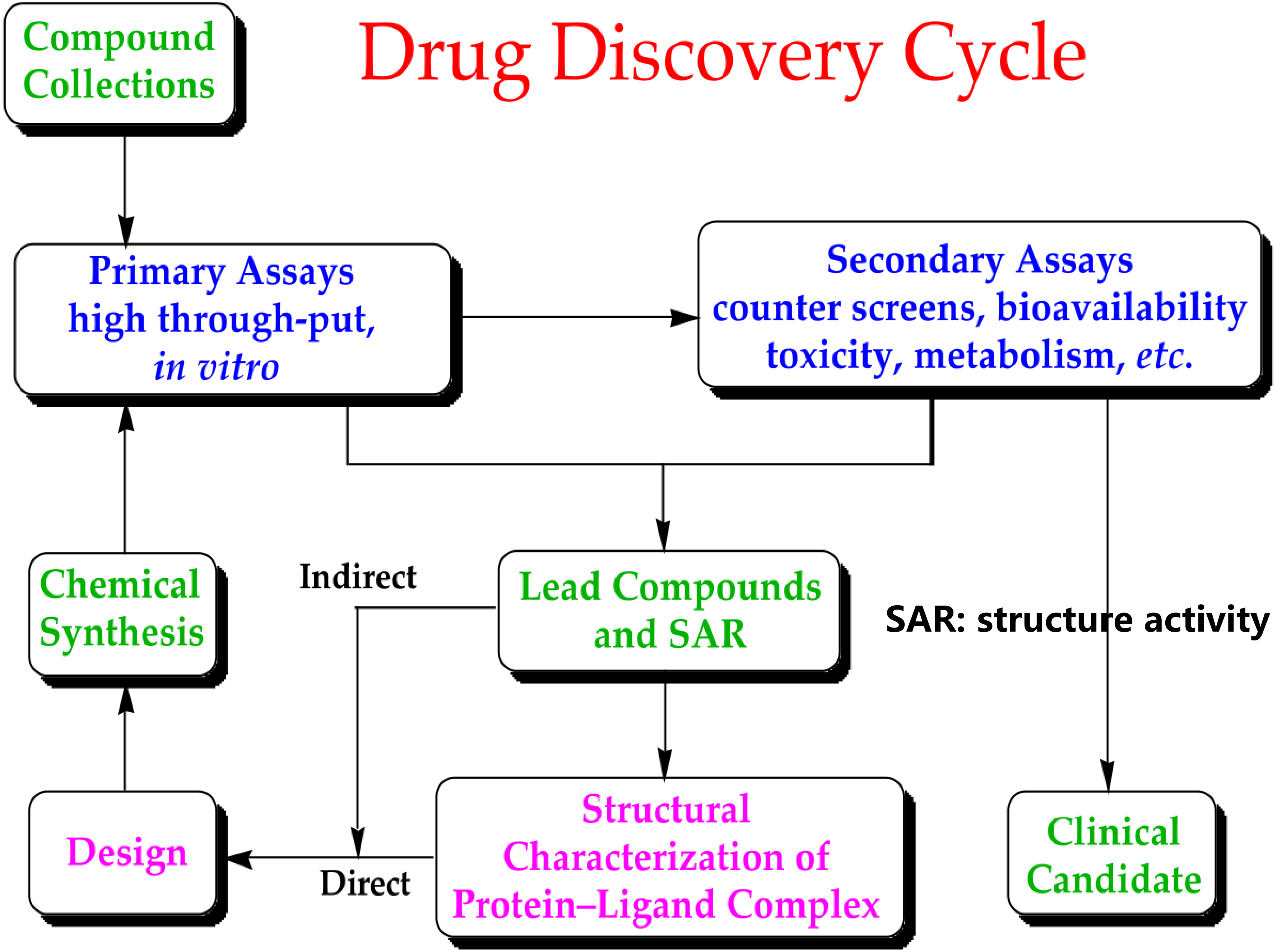
NMEs: Include all peptides and oligonucleotides of up to 40 amino acids in length.
BLAs: Protein-based candidates.

Nature Reviews Drug Discovery, 2020, 19, 79

Computer-Aided Drug Design (CADD)
Artificial Intelligence-Driven Drug Design (AIDD)



Drug Discovery Cycle



SAR: structure activity relationship

AI for Drug Development

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Lots of things happening: time for critical evaluation of where we are

nature

SPOTLIGHT | 30 May 2018

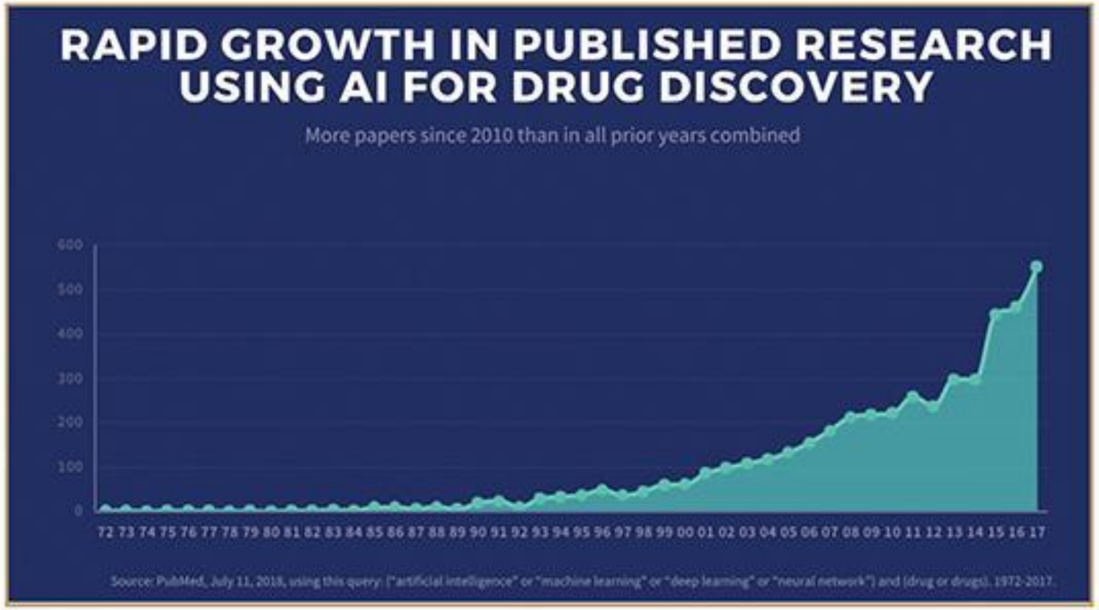
How artificial intelligence is changing drug discovery

Machine learning and other technologies are expected to make the hunt for new pharmaceuticals quicker, cheaper and more effective.

Pfizer — immuno-oncology drugs

Sanofi, Exscientia — metabolic-disease therapies

Roche, Genentech (IN BUSINESS FOR LIFE) — cancer treatments



Here come robots…

12 June 2007

"Adam" to carry out the entire scientific process on its own: formulating hypotheses, designing and running experiments, analyzing data, and deciding which experiments to run next.

Adam

- Identify the function of a gene
- Generate hypothesis which genes are key enzyme
- Validate by itself (9/10 are right)

Eva

- Test compound (4/100 are active)
- Discover Compounds against malaria parasites

Drug Discovery and Development Timeline

Mfg.: Manufacturing

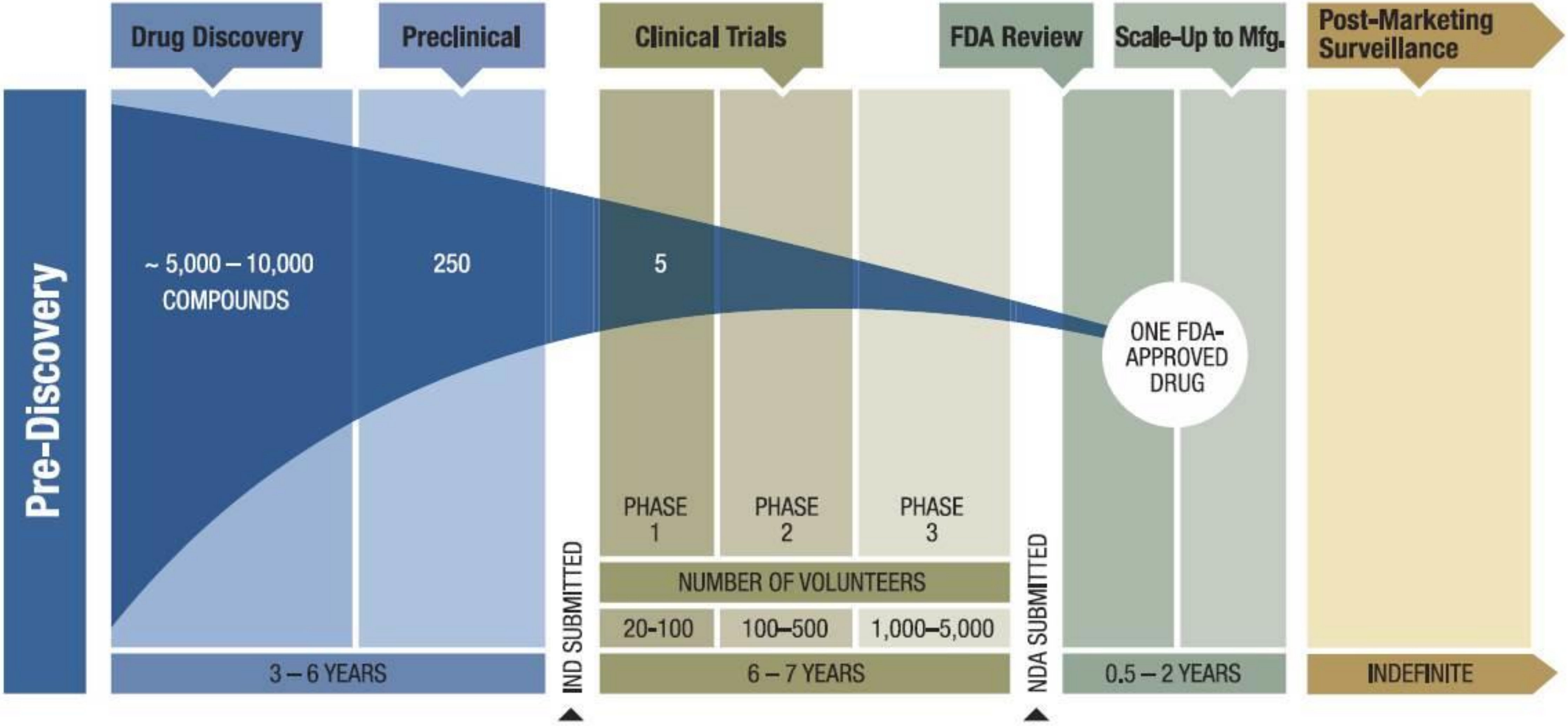
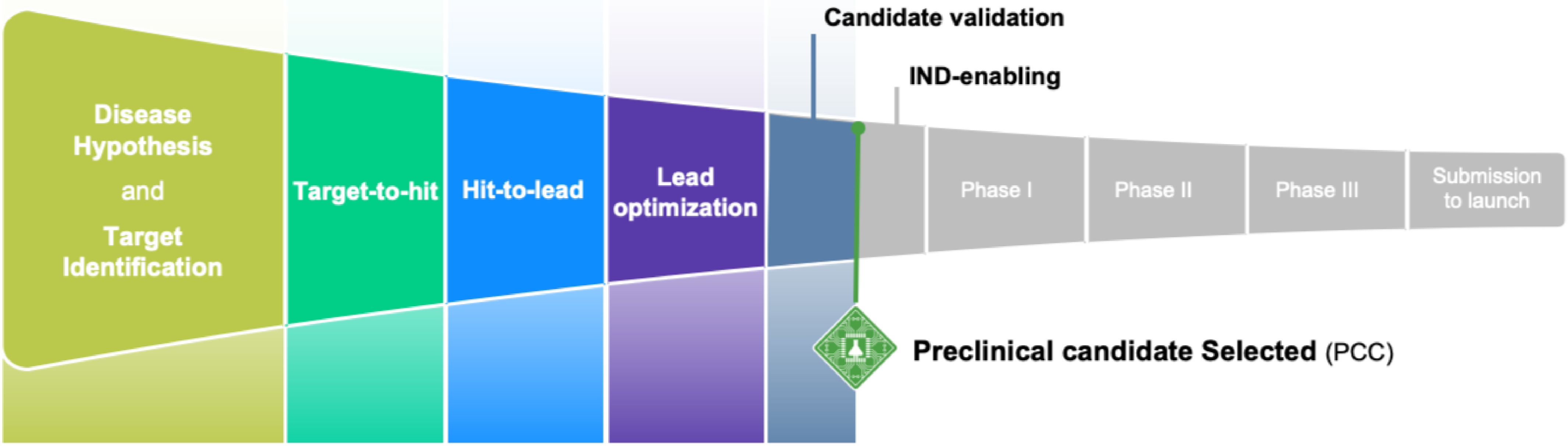


Figure adapted from: https://friendsofcancerresearch.org/cancer-drug-approval

Our AI system has revealed a novel wide-indication target, and a corresponding drug candidate in under 18 months and at roughly 1/10th of the typical cost associated with similar programs.

Traditional Approach

Usually performed in academia using public funds and may take decades	1 year	1.5 years	2 years	Cycle time
	$ 94M	$ 166M	$ 414M	Cost per launch (capitalized) in 2010, and increasing

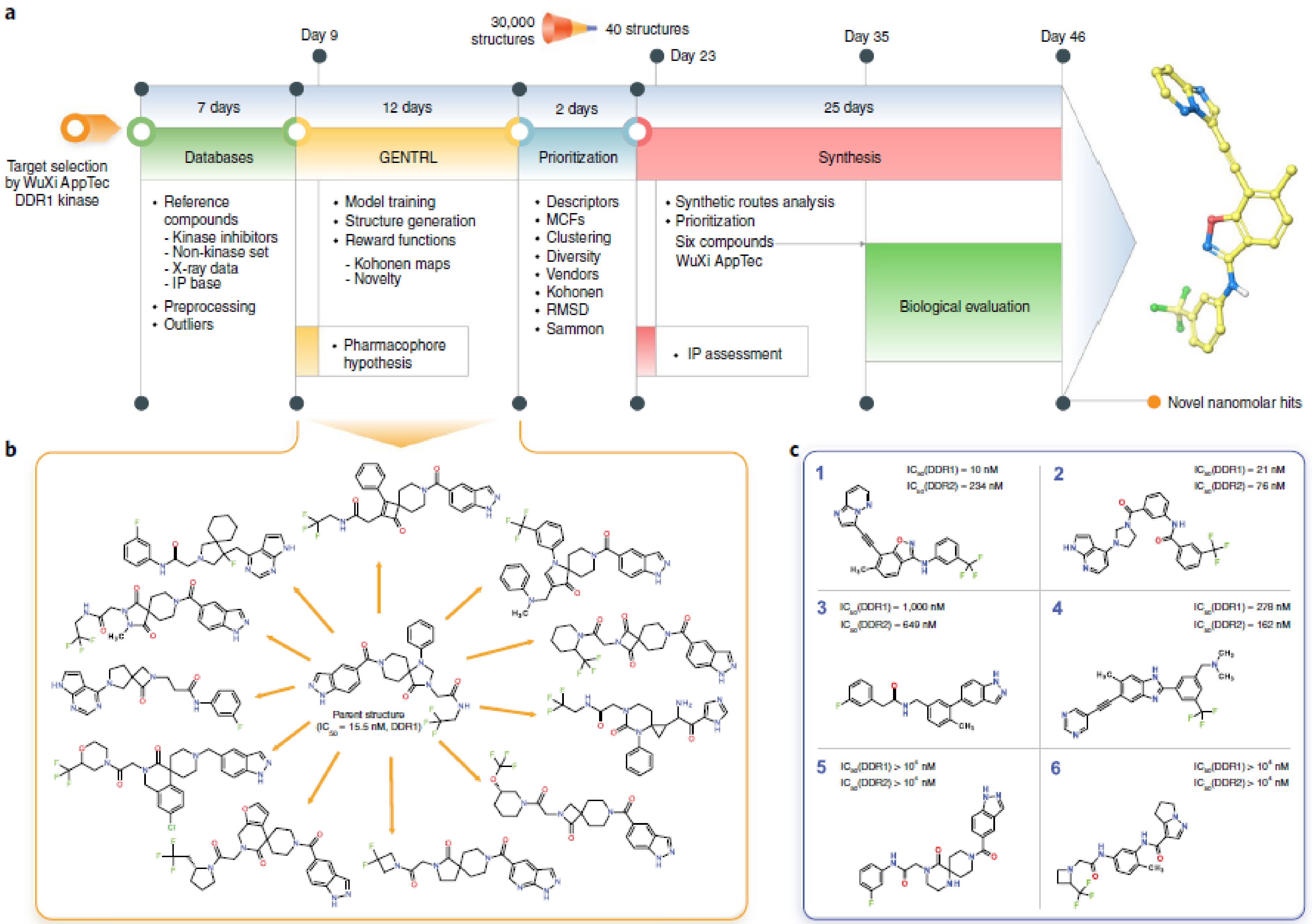


AI	1 week	2 months	4 months	11 months	Cycle time
	< $ 50k	$ 200k	$ 400k	$ 2M	Cost at each stage

Insilico Medicine Approach

https://insilico.com/

Discovery Development



Generative Tensorial Reinforcement Learning (GENTRL) model

Deep learning enables rapid identification of potent DDR1 kinase inhibitors

Nature Biotechnology, 2019,1038–1040

AI can Help in Many other areas of Drug Discovery

https://insilico.com/

Drug Target

What is a drug target?

- **Therapeutic Targets(治疗靶点)：**

a protein or nucleic acid whose activity can be modified by an external stimulus.

- **Surrogate Targets(替代靶点)：**

remedy the unfavorable prognosis, rather than to correct the physiological abnormality per se.

- **Drug Targets(药物靶点):**

A drug target is a molecule in the body, usually a protein, that is intrinsically associated with a particular disease process and that could be addressed by a drug to produce a desired therapeutic effect.

External Stimulus:

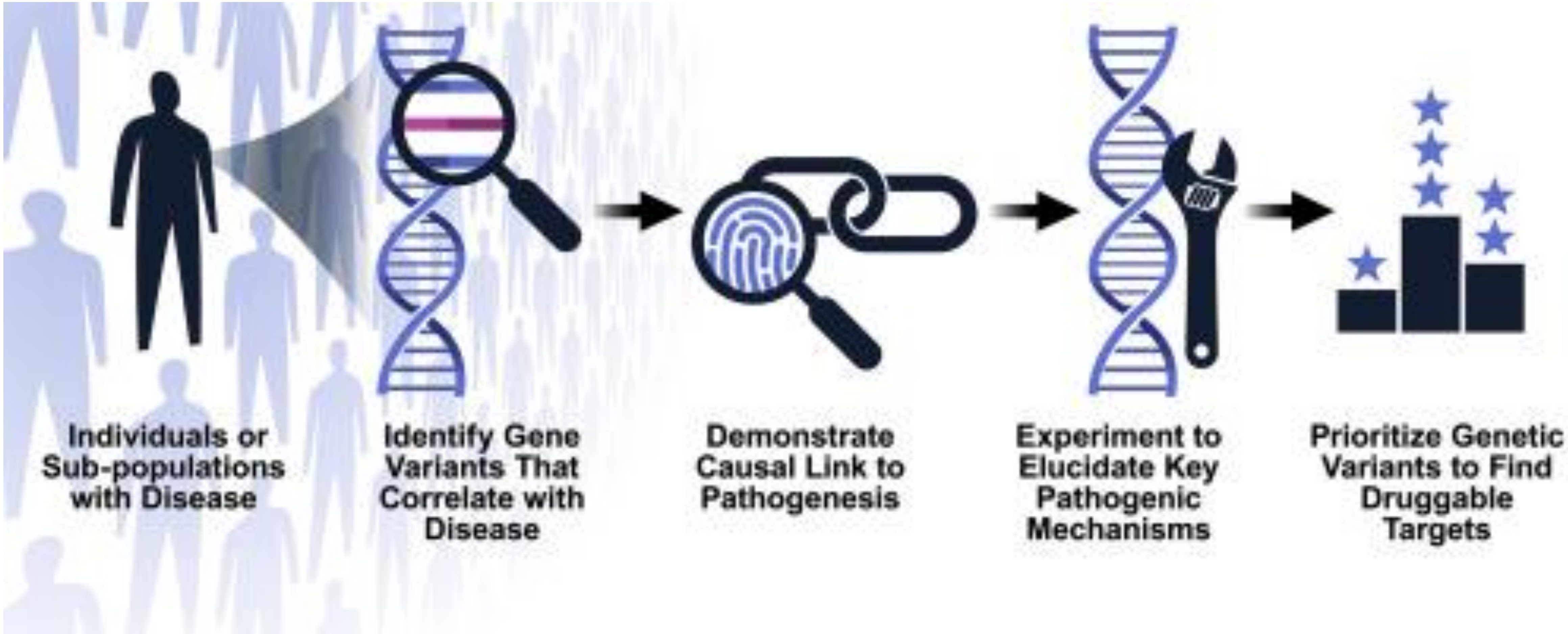
- **Reversible Binding:**

Traditional reversible drugs are in equilibrium with their target-continually binding, unbinding, & rebinding

- **Covalent Binding (can silence proteins):**

Covalent irreversible drugs bind specifically to a drug target and form a precisely directed, permanent bond with their target.

How to define a potential drug target?



How to design drugs against a target?

Traditional Chinese Medicine

Modern Drug Development

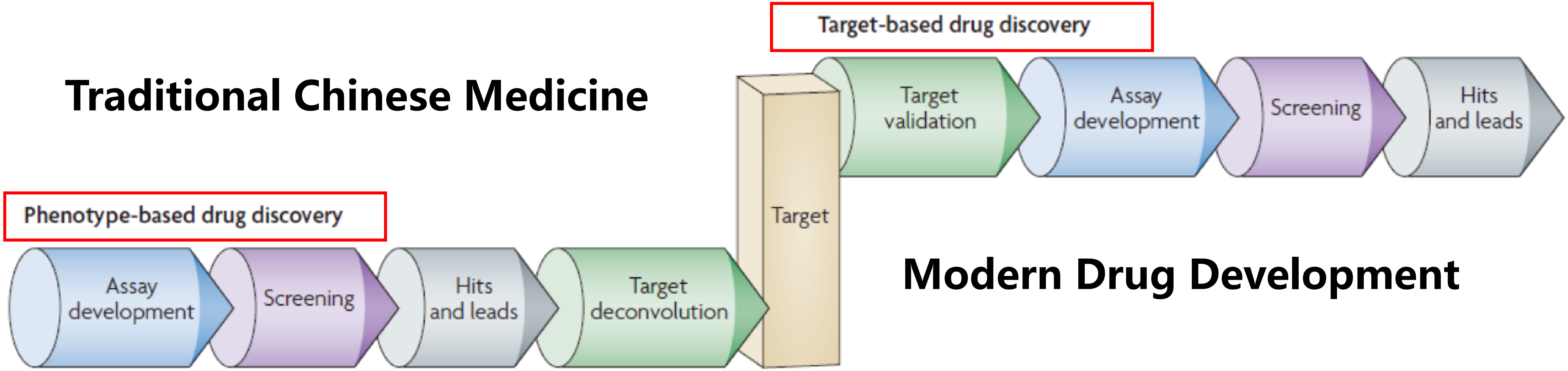


Figure 1 | **Phenotype-based versus target-based drug discovery.** The diagram illustrates the early phase of drug discovery, in which the aim is to identify target and lead molecules. In the phenotype-based approach, lead molecules are obtained first, followed by target deconvolution to identify the molecular targets that underlie the observed phenotypic effects. In the target-based approach, molecular targets are identified and validated before lead discovery starts; assays and screens are then used to find a lead.

Drug target databases

Therapeutic Target Database (TTD)

TTD

中国；
浙大朱峰

Content	
Description	Drug target database
Contact	
Laboratory	Innovative Drug Research and Bioinformatics Group (IDRB) Bioinformatics and Drug Design Group (BIDD)
Primary citation	PMID 31691823
Release date	11 Nov, 2019
Access	
Website	http://db.idrblab.net/ttd/
Miscellaneous	
License	Free access
Version	7.1.01

Constructed based:
(i) target-regulating microRNAs and transcription factors;
(ii) target-interacting proteins;
(iii) patented agents and their targets;

BindingDB

BindingDB

美国

Content	
Description	Chemical database
Data types captured	Molecules with drug-like properties and biological activity
Contact	
Authors	Michael Gilson, Team Leader
Primary citation	PMID 17145705
Release date	1995
Access	
Website	BindingDB
Download URL	Downloads
Miscellaneous	
License	The BindingDB data is made available on a Creative Commons Attribution-Share Alike 3.0 Unported License

focusing chiefly on the interactions of proteins considered to be candidate drug-targets with ligands that are small, drug-like molecules.

DrugBank

DRUGBANK

加拿大

Comprehensive database

Content	
Description	Drug database
Data types captured	Chemical structures, small molecule drugs, biotech drugs, drug targets, drug transporters, drug target sequences, drug target SNPs, drug metabolites, drug descriptions, disease associations, dosage data, food and drug interactions, adverse drug reactions, pharmacology, mechanisms of action, drug metabolism, chemical synthesis, patent and pricing data, chemical properties, nomenclature, synonyms, chemical taxonomy, drug NMR spectra, drug GC-MS spectra, drug LC-MS spectra
Contact	
Research center	University of Alberta and The Metabolomics Innovation Centre, Alberta, Canada
Access	
Website	www.drugbank.com
Download URL	www.drugbank.ca/downloads

Computational drug target identification

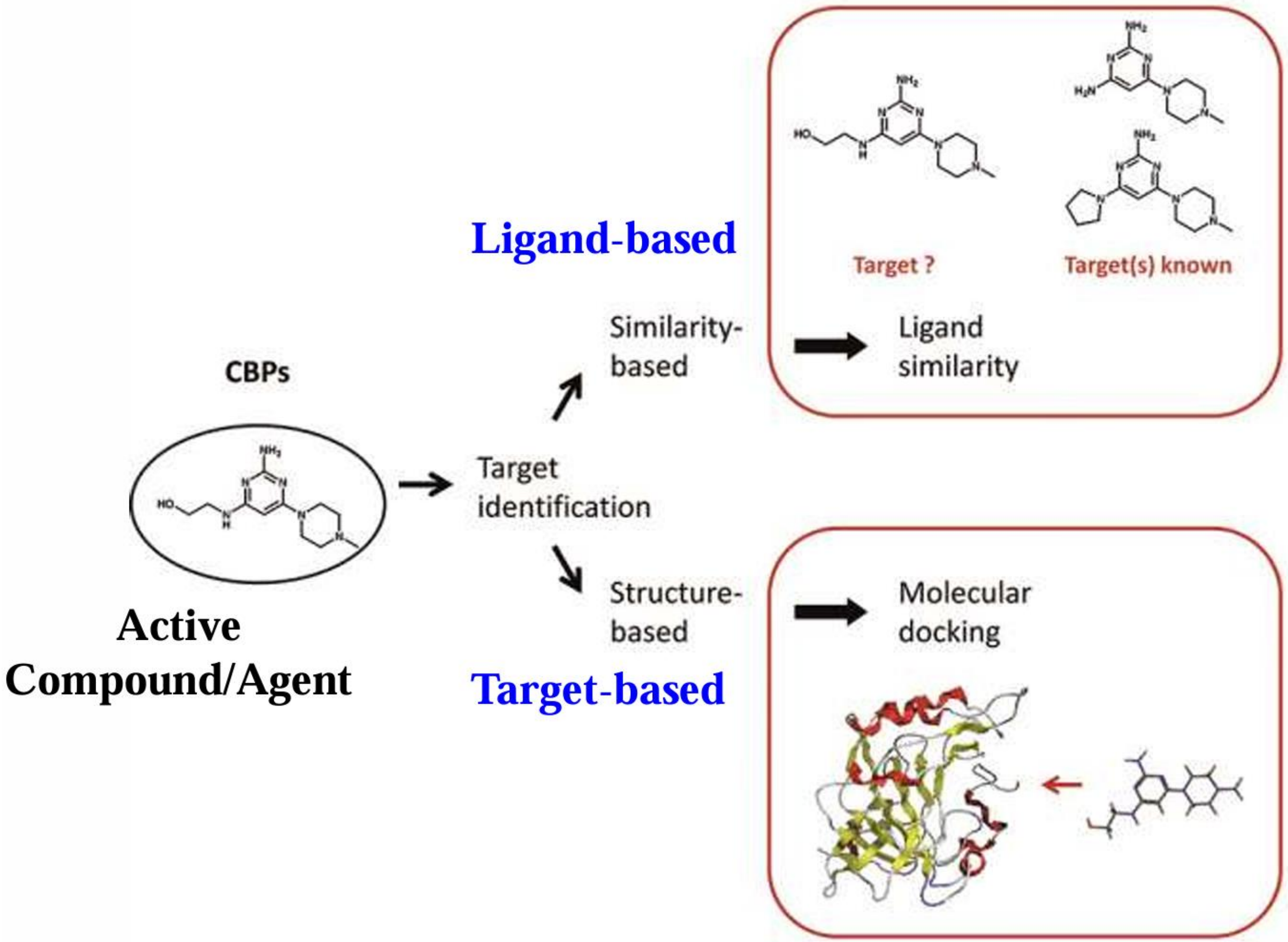


Figure from:https://analyticalscience.wiley.com/

Pharmacophore-based

- 形状匹配；
- 物理化学性质匹配；
- 结合能量最低；

Drug Discovery Today, 2008, 13:23-9

Drug-Target Interaction

Predicting and quantifying drug-target binding

Virtual screening
- Large chem space
- Hypothesis-generating

Bioassays
- High functional relevance
- Prone to artifacts

DSF
- Fast assay development
- Indirect readout

SPR/BLI
- Kd, stoichiometry
- Immobilize protein

Ligand NMR
- Widely applicable
- Slow, material-intensive

Mass spectrometry
- Site-directed Tethering
- Native MS in development

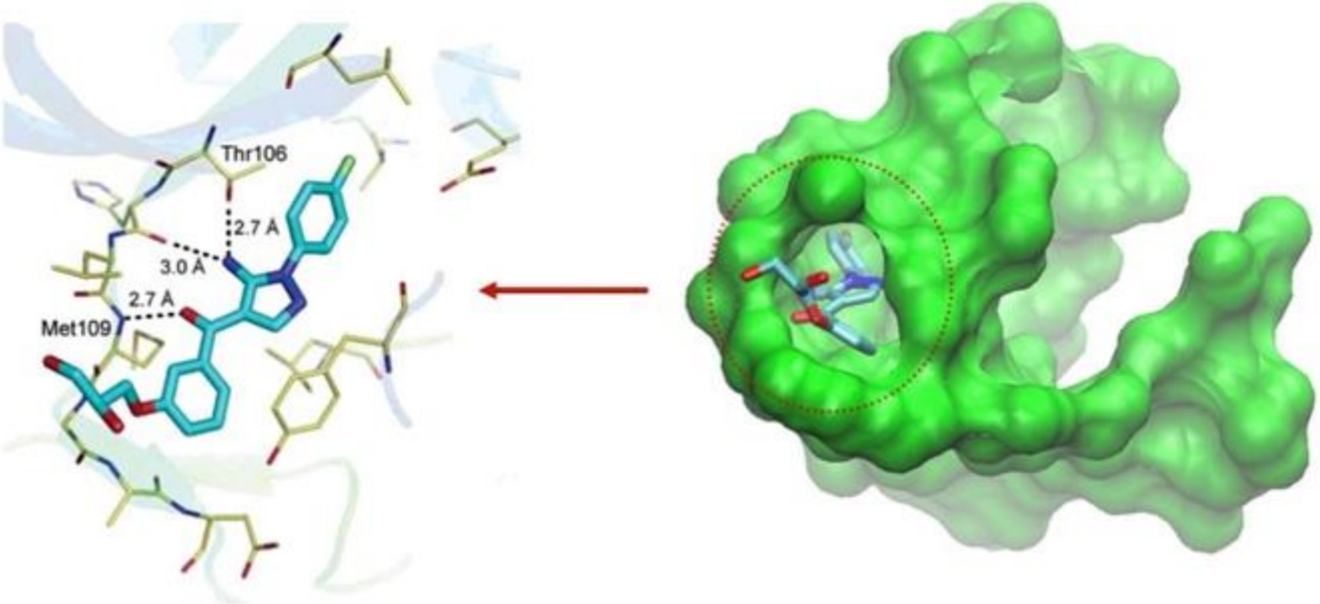
Protein NMR
- Structural information
- Size and material limitations

ITC
- Label free, Kd and stoichiometry
- Material, solubility limitations

Crystallography
- Label free, high resolution
- Prone to false -ves

Relative throughput

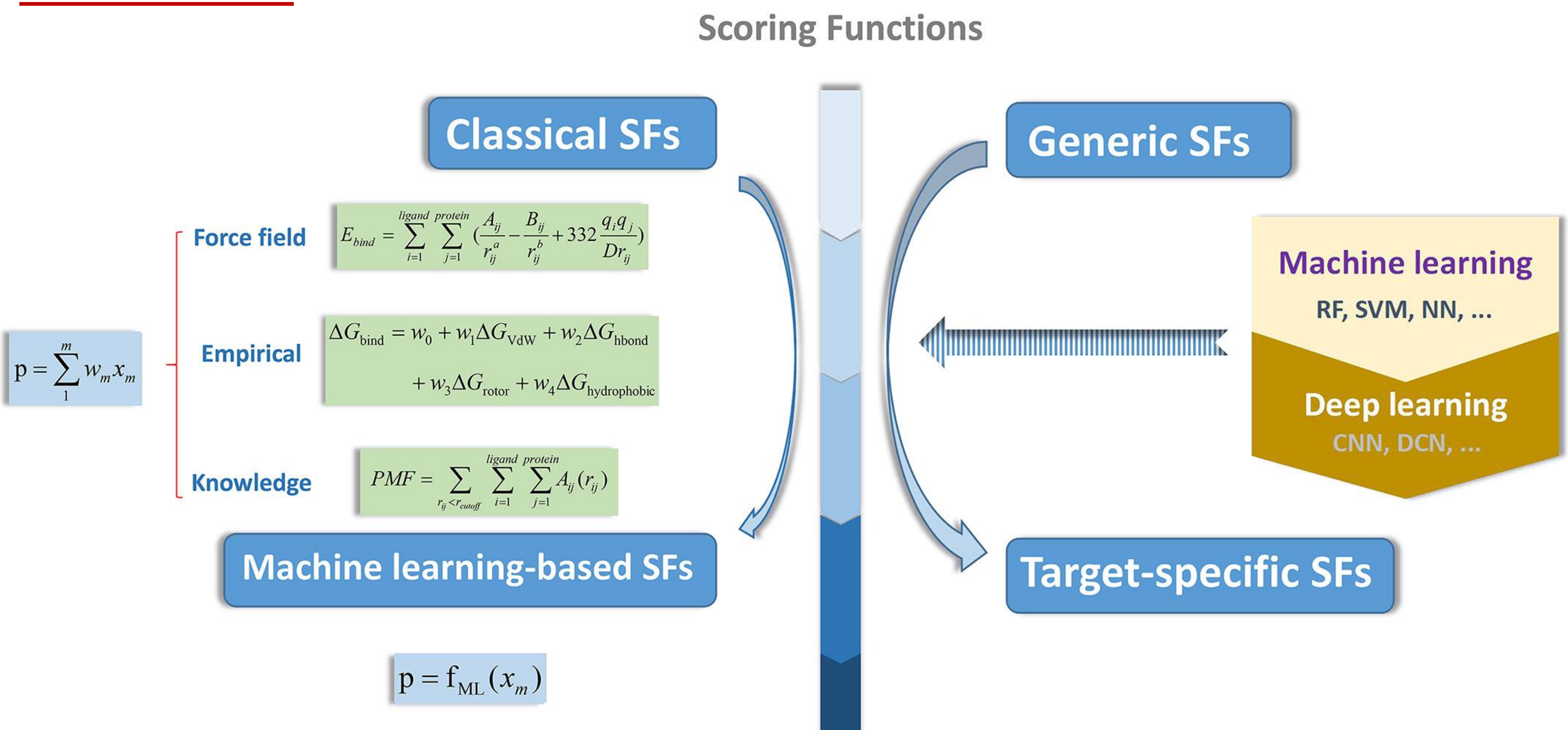
Relative information content



$$K_d = \frac{[R][L]}{[R \cdot L]}$$

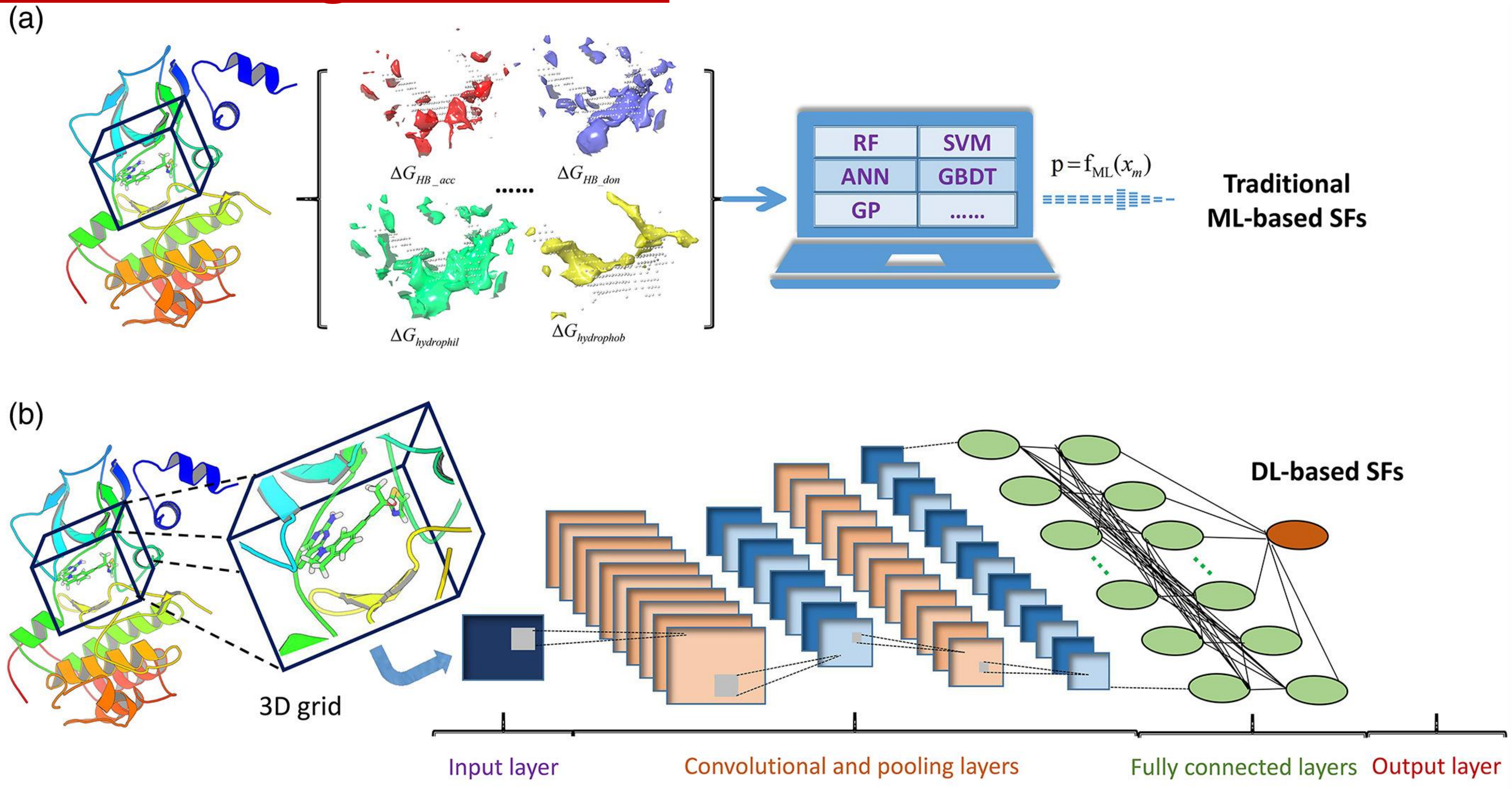
Strength of binding: R + L ←→ R•L

Estimating binding affinity by using scoring functions



WIREs Comput Mol Sci. 2020;10:e1429

Typical workflows for the development of AI-based scoring functions



WIREs Comput Mol Sci. 2020;10:e1429

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Drug Action Mechanisms

Central dogma of molecular biology vs drug design

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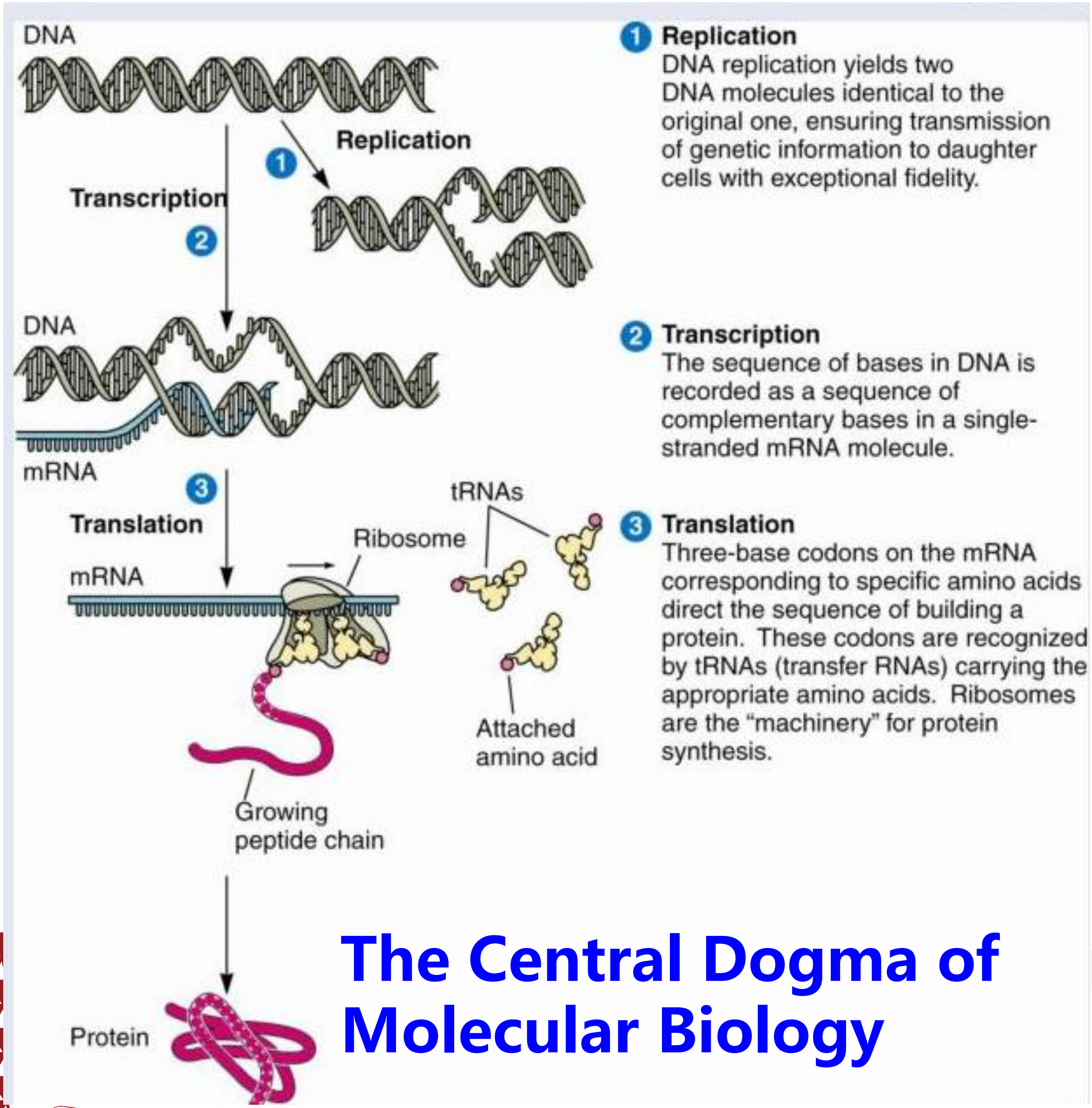
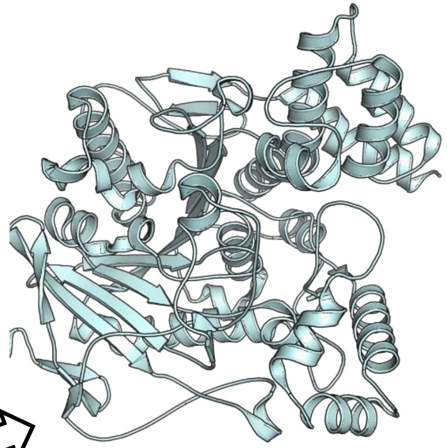
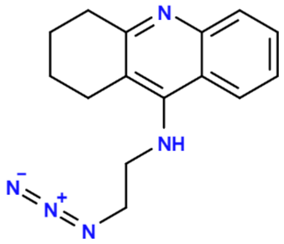


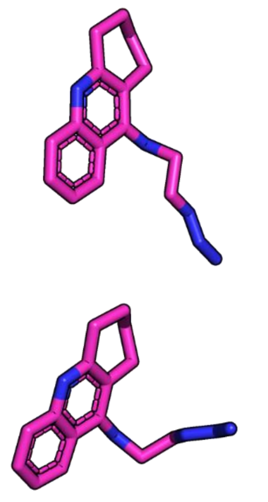
The Central Dogma of Molecular Biology



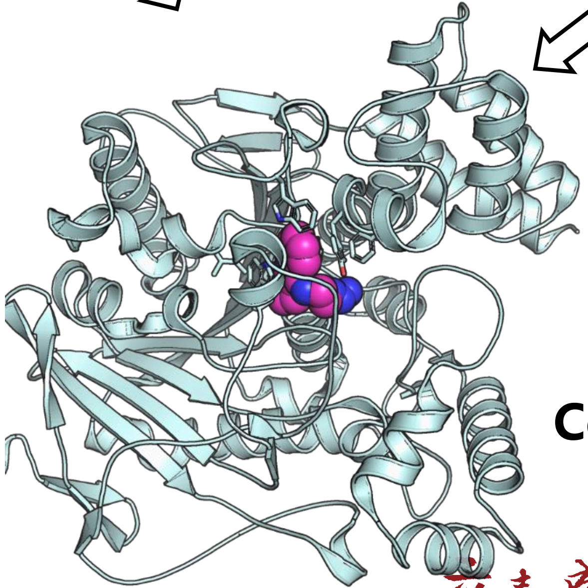
3D Structure

Drug





$\Delta G_{\text{binding}}$



Protein-Drug Recognition

Conformation Prediction

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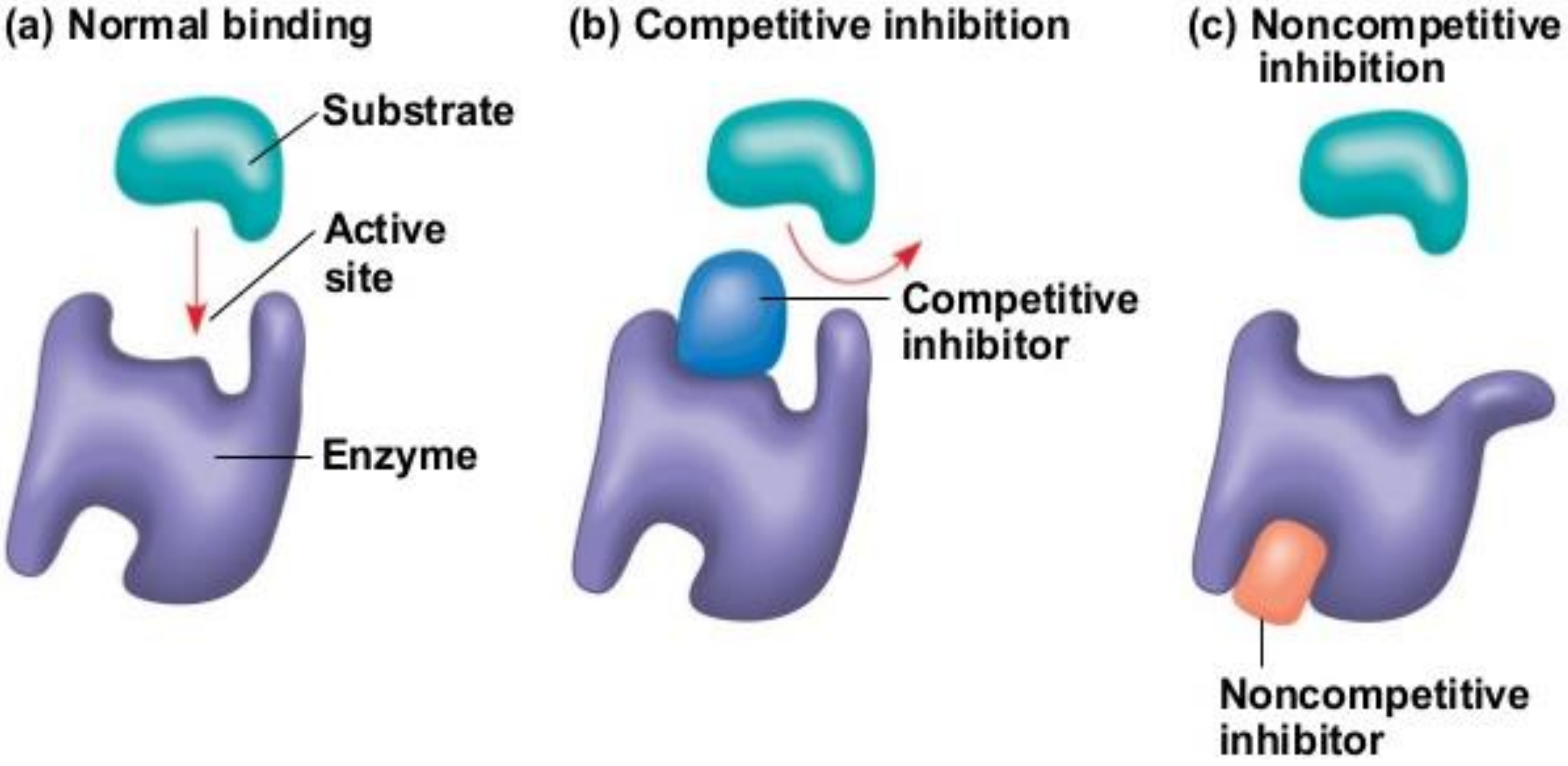
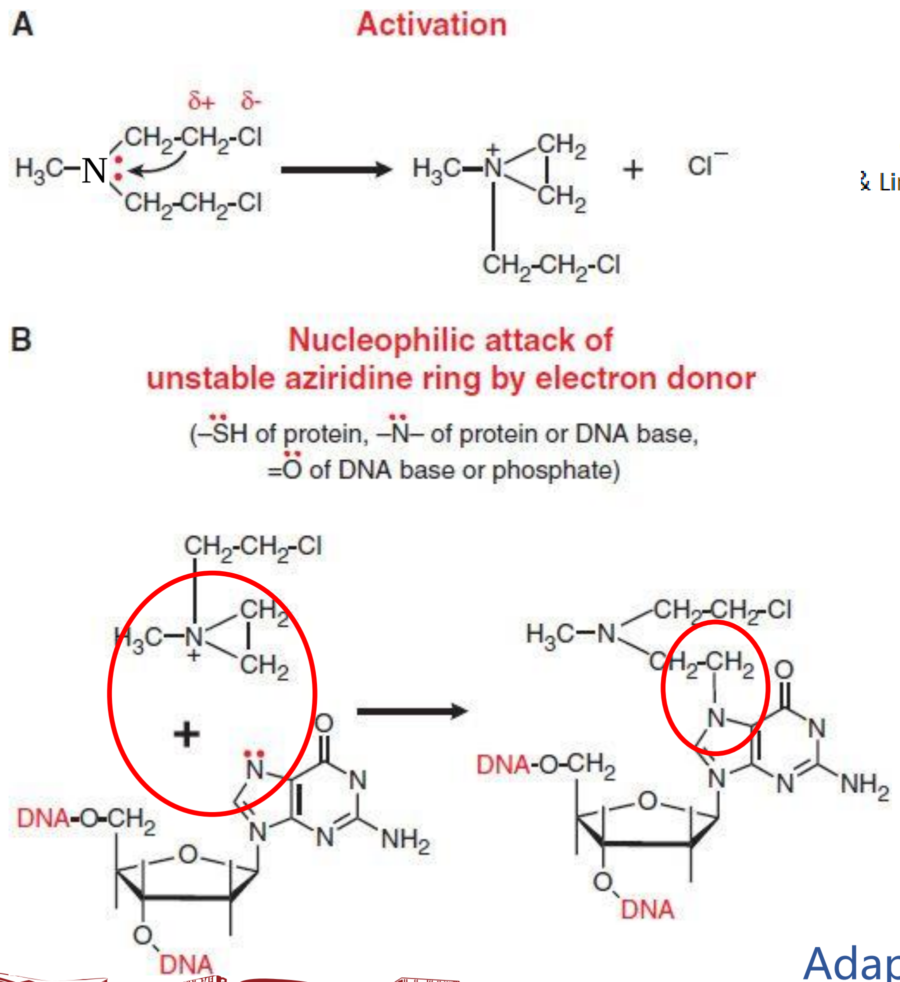
Drug action mechanisms

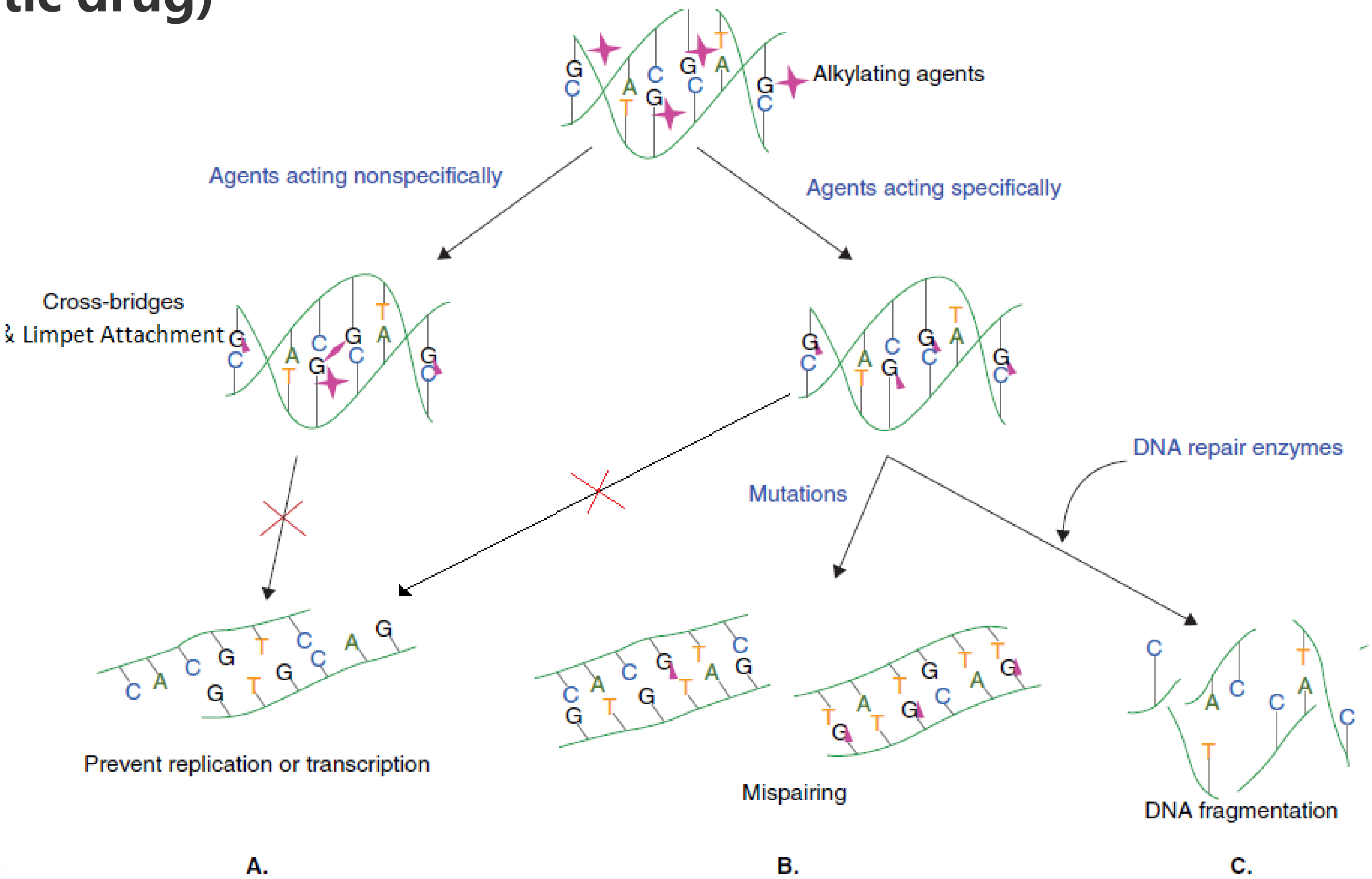
Enzyme Inhibitors



Drug action mechanisms

Cytotoxic drug (chemotherapeutic drug)
细胞毒性药物：e.g. 烷化剂





阻碍DNA链解旋

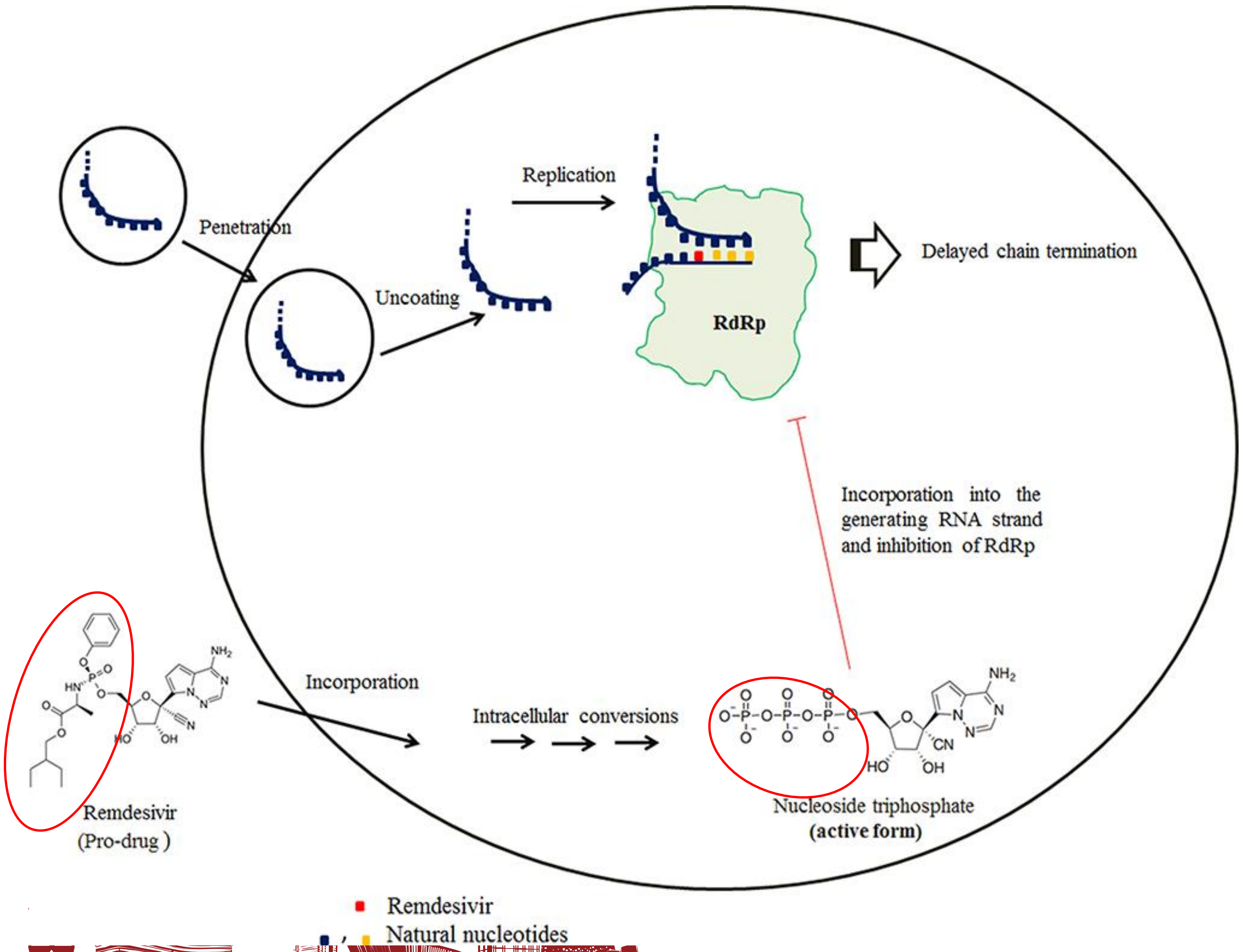
引起碱基对错配，从而诱发突变

过度激活DNA修复酶，导致DNA断裂

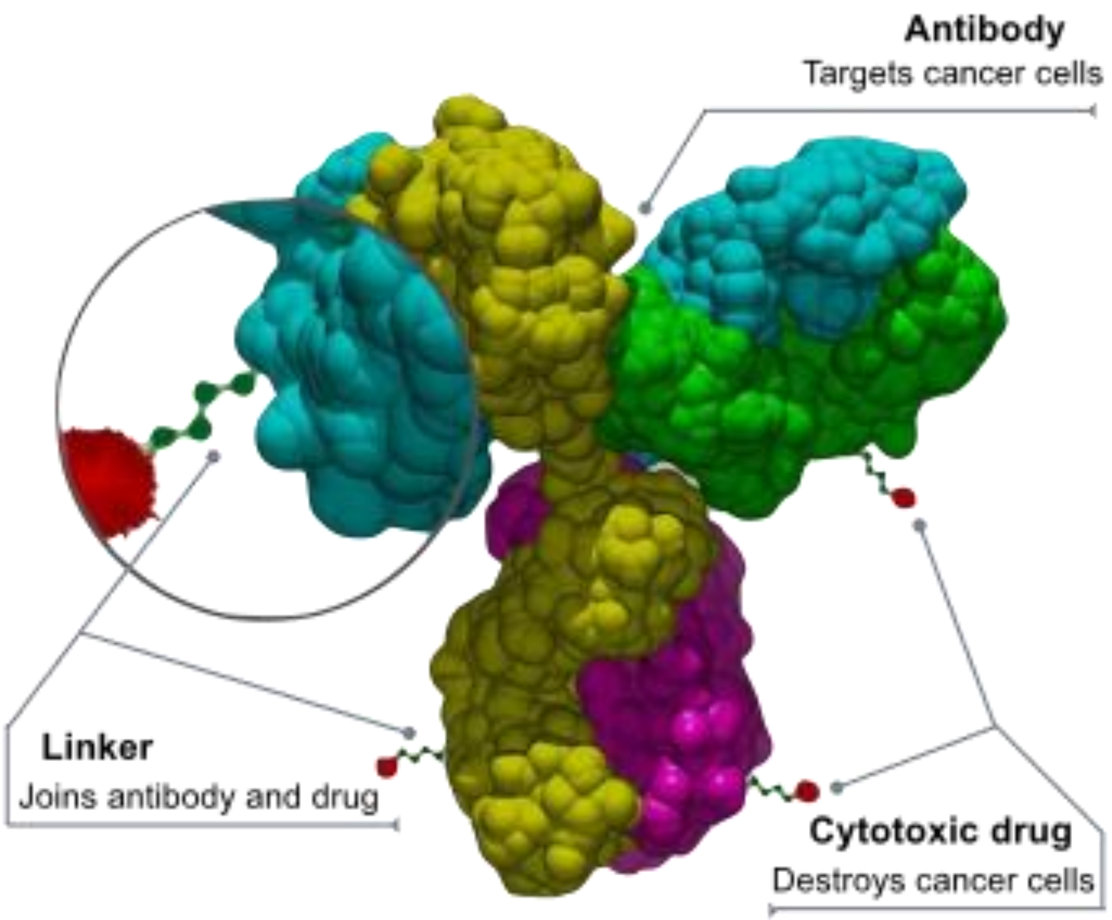
Adapted from Ralhan and Kaur, 2007

Drug action mechanisms

RNA drug Remdesivir

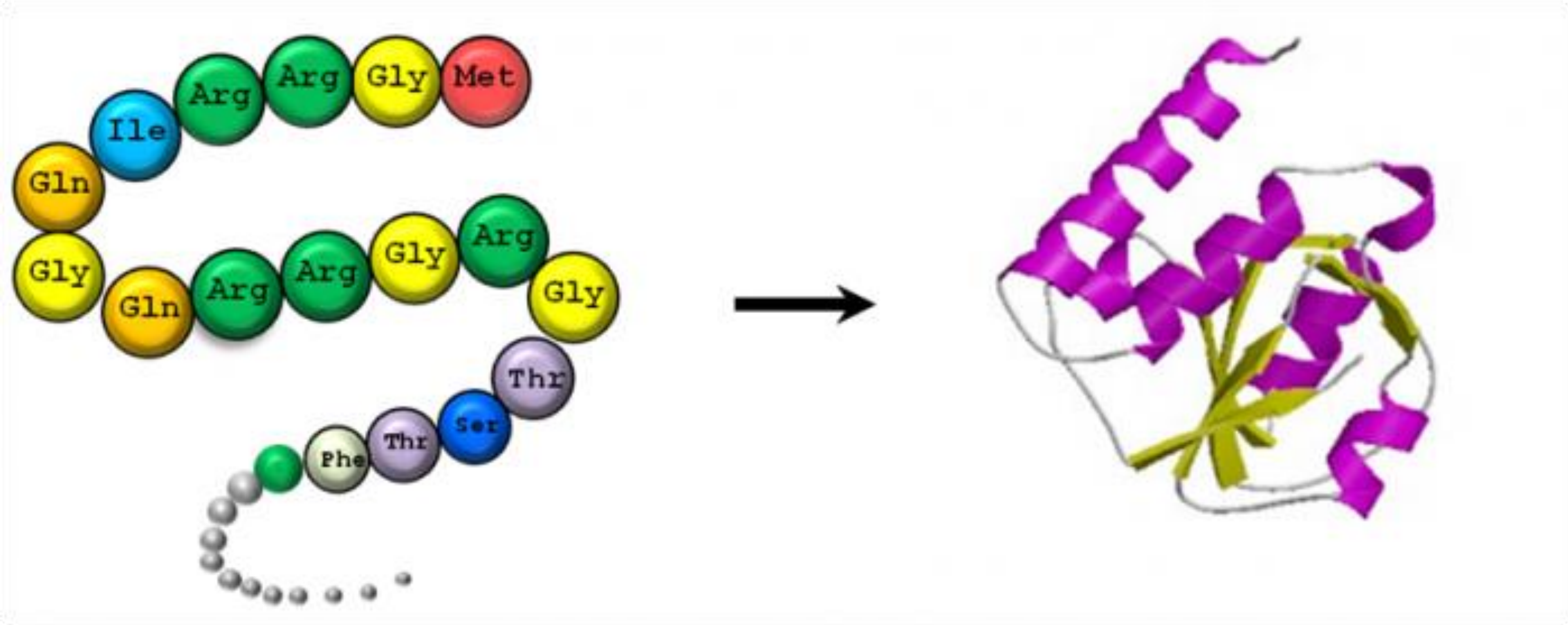


Antibody-drug conjugate

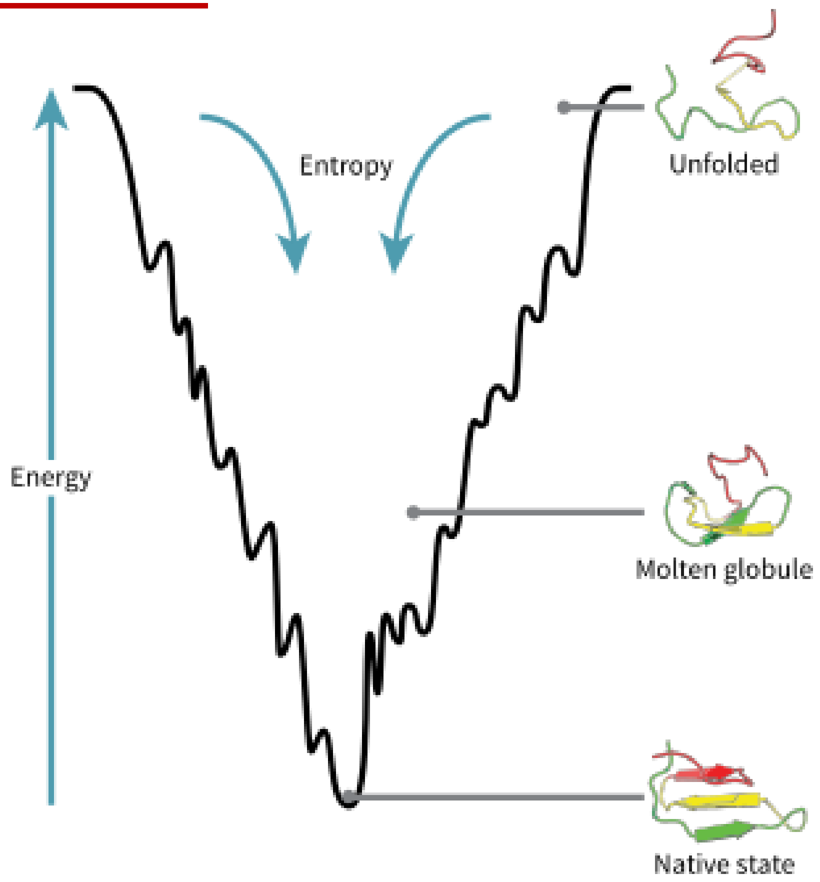


Protein Structure

Protein structure prediction

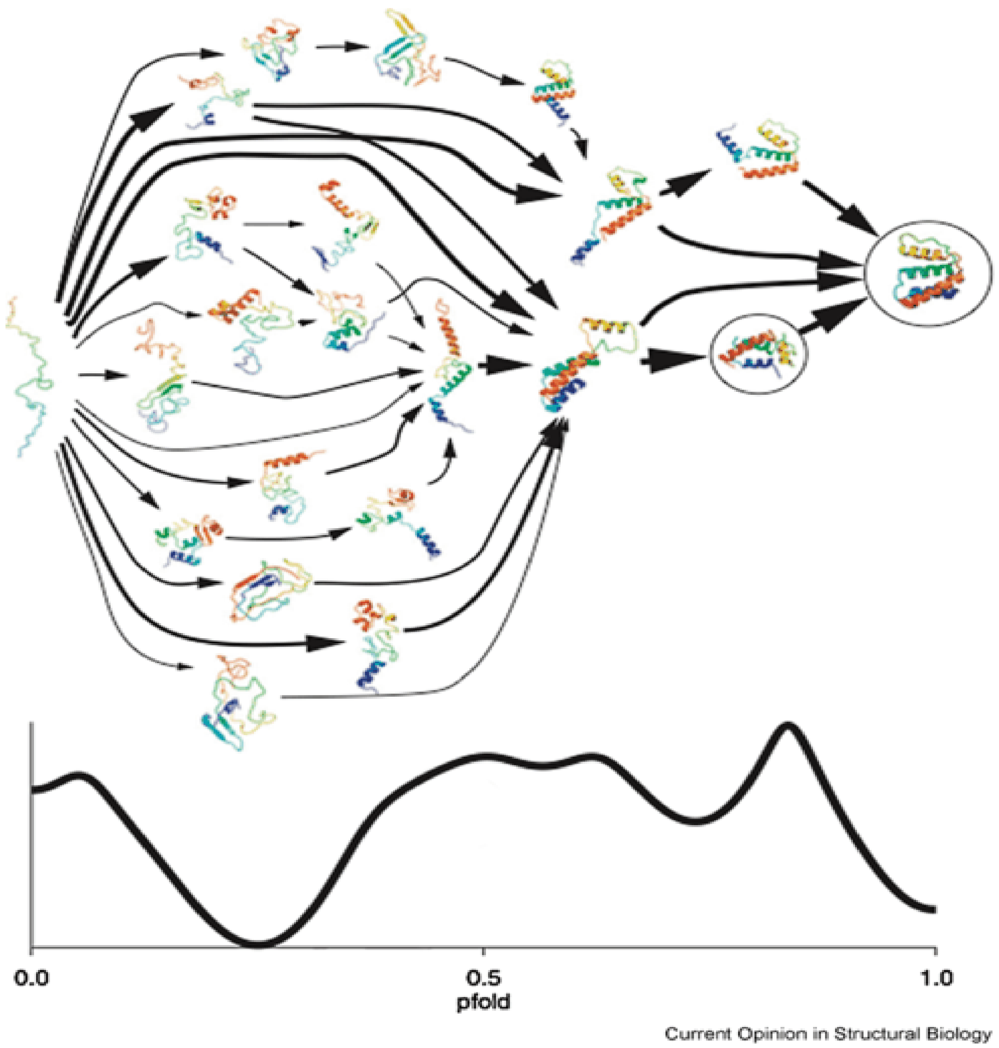


Protein folding: energy landscape and folding funnel



$$\Delta G = \Delta H^{\circ}_{sys} - T\Delta S_{system}$$

Enthalpy (焓) Entropy (熵)



Curr. Opin. Struct. Biol. **2004**,14:70–75

The boom of AI in protein structure prediction

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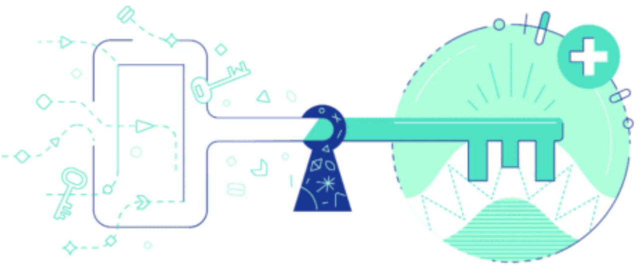
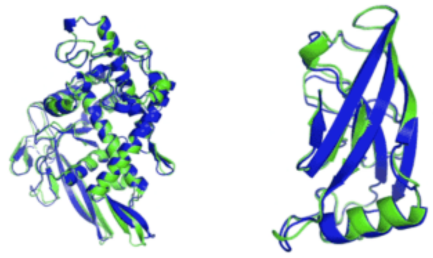
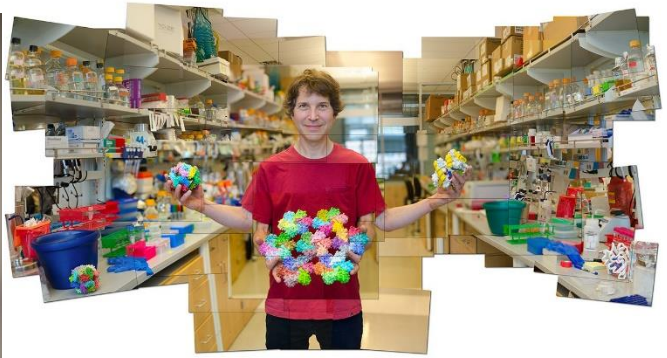
DeepMind







Baker Lab

nature https://doi.org/10.1038/s41586-021-03819-2

Accelerated Article Preview

Highly accurate protein structure prediction with AlphaFold

2021 July 16

nature https://doi.org/10.1038/s41586-021-03828-1

Accelerated Article Preview

Highly accurate protein structure prediction for the human proteome

2021 July 22

Figure from: https://metacurate.io/

Science RESEARCH ARTICLES

Cite as: M. Baek *et al.*, *Science* 10.1126/science.abj8754 (2021).

Accurate prediction of protein structures and interactions using a three-track neural network

doi: https://doi.org/10.1101/2021.08.15.456425; this version posted August 15, 2021. The copyright holder fc
ertified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpe
available under aCC-BY-NC-ND 4.0 International license.

ColabFold - Making protein folding accessible to all

Milot Mirdita,[1,*] Sergey Ovchinnikov,[2,3,*] and Martin Steinegger[4,5,*]

Powered by AlphaFold2 and RoseTTAFold combined with a fast multiple sequence alignment generation stage using MMseqs2.

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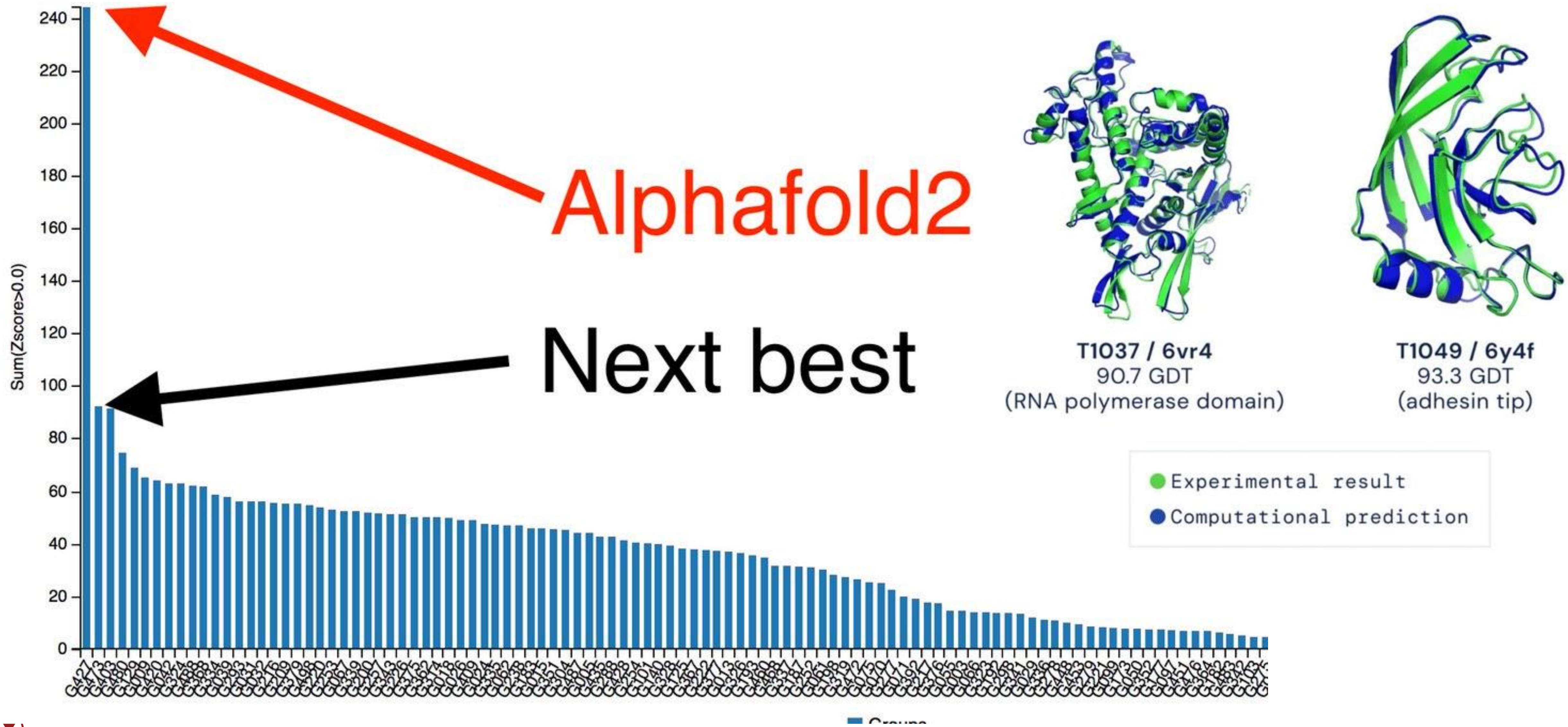
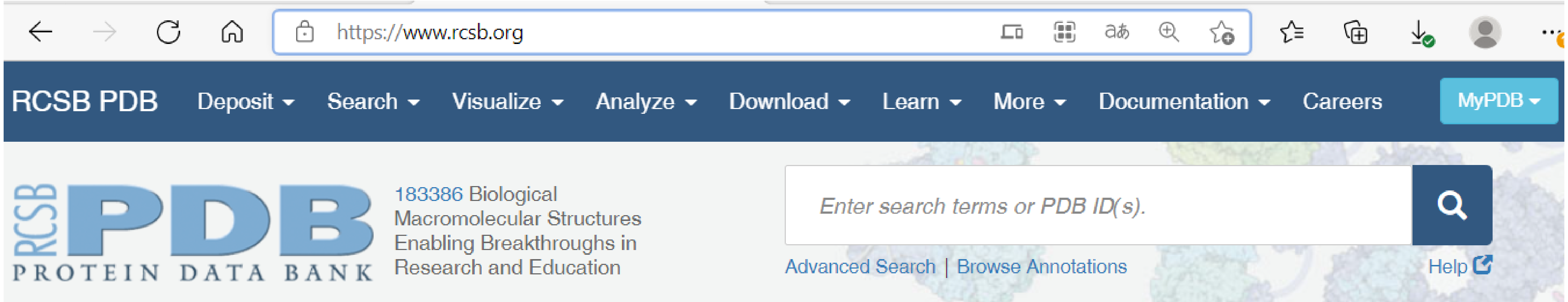
AlphaFold: AI-based protein structure prediction



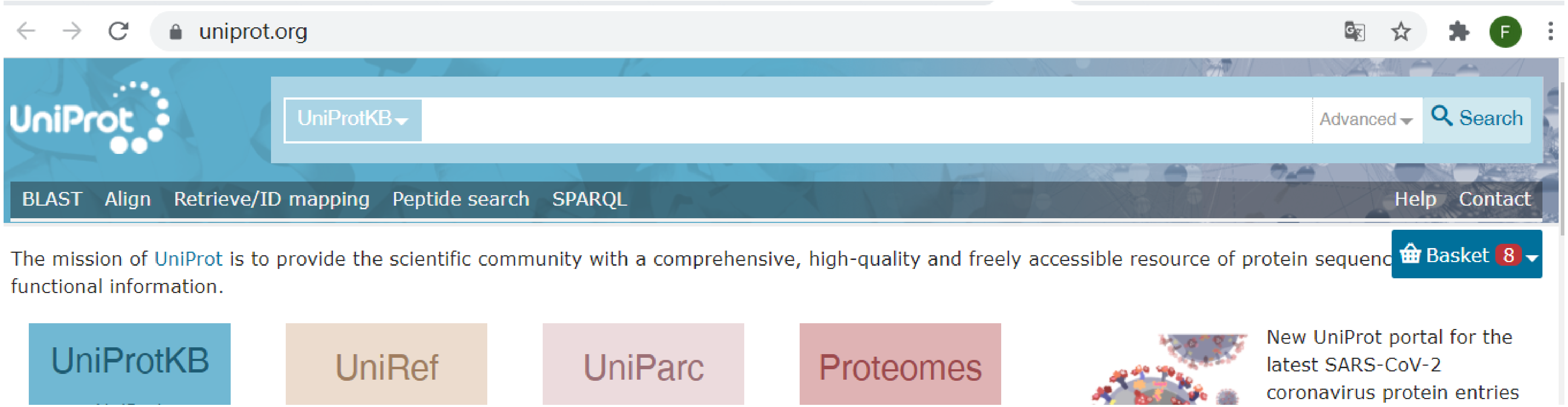
蛋白质结构数据库

https://www.rcsb.org/



蛋白序列数据库

https://www.uniprot.org/



Challenges there, waiting for climbers, to be conquered…

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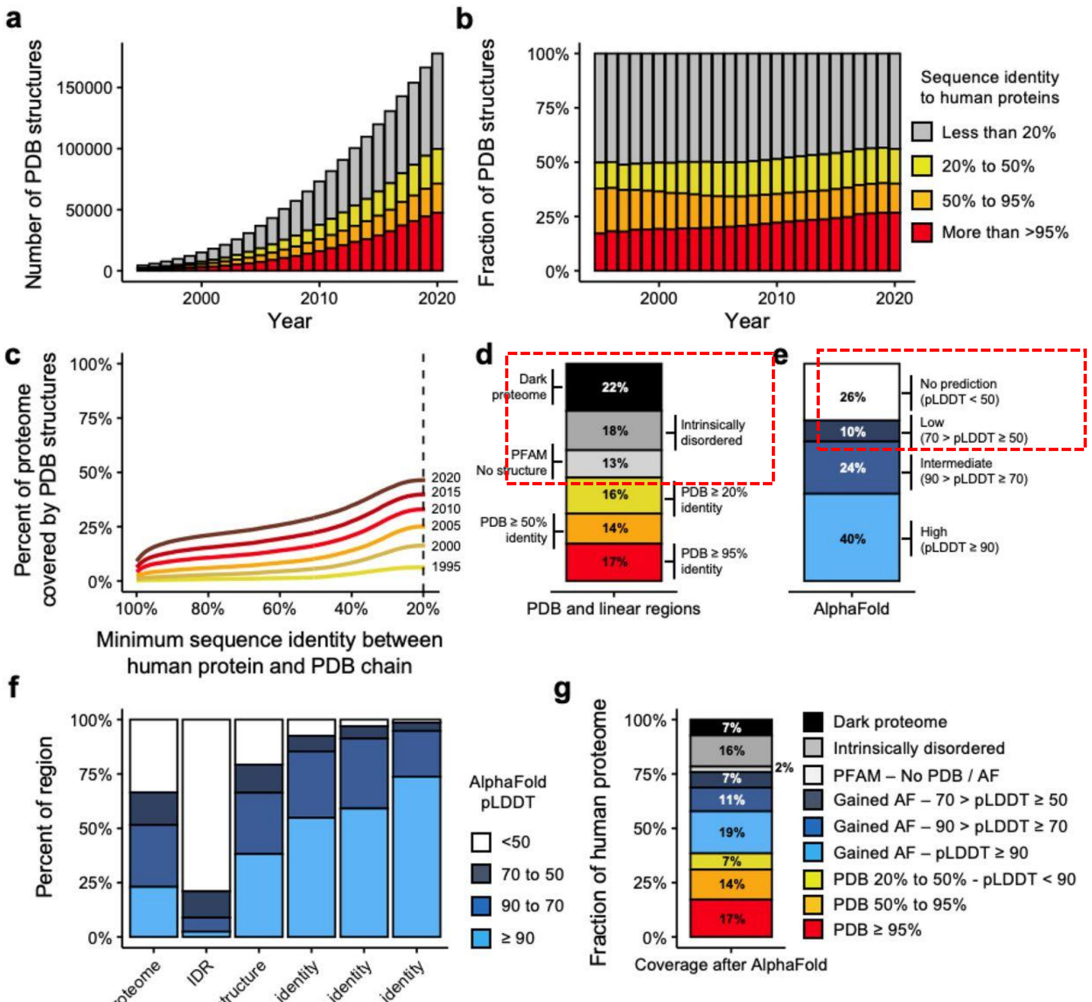
- Number of protein structures is greatly increasing
- Human proteome is highly covered
- Large portion of the human proteome that lacks structure coverage

Current coverage of the human proteome



BioRxiv: doi: https://doi.org/10.1101/2021.08.03.454980

- Contribution of human proteins has also increased

Dark proteome: neither structured or predicted to be disordered

- Large portion of challengeable proteins (dark proteome, disordered proteins…)
- Large portion of low-confidential prediction
- We have ~30% proteome to work with

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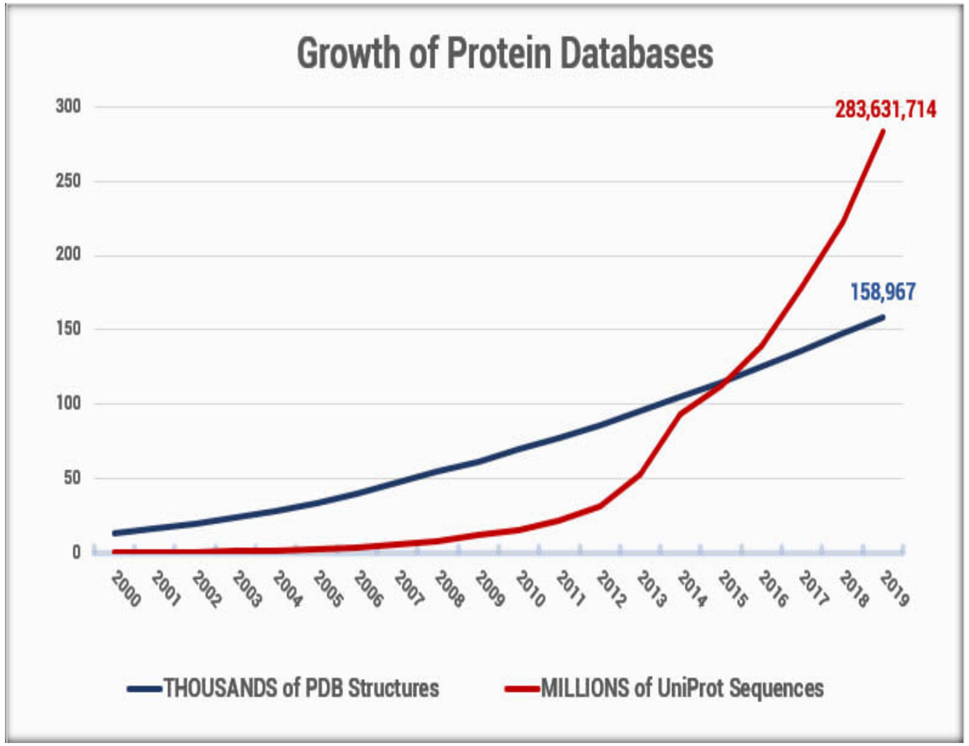
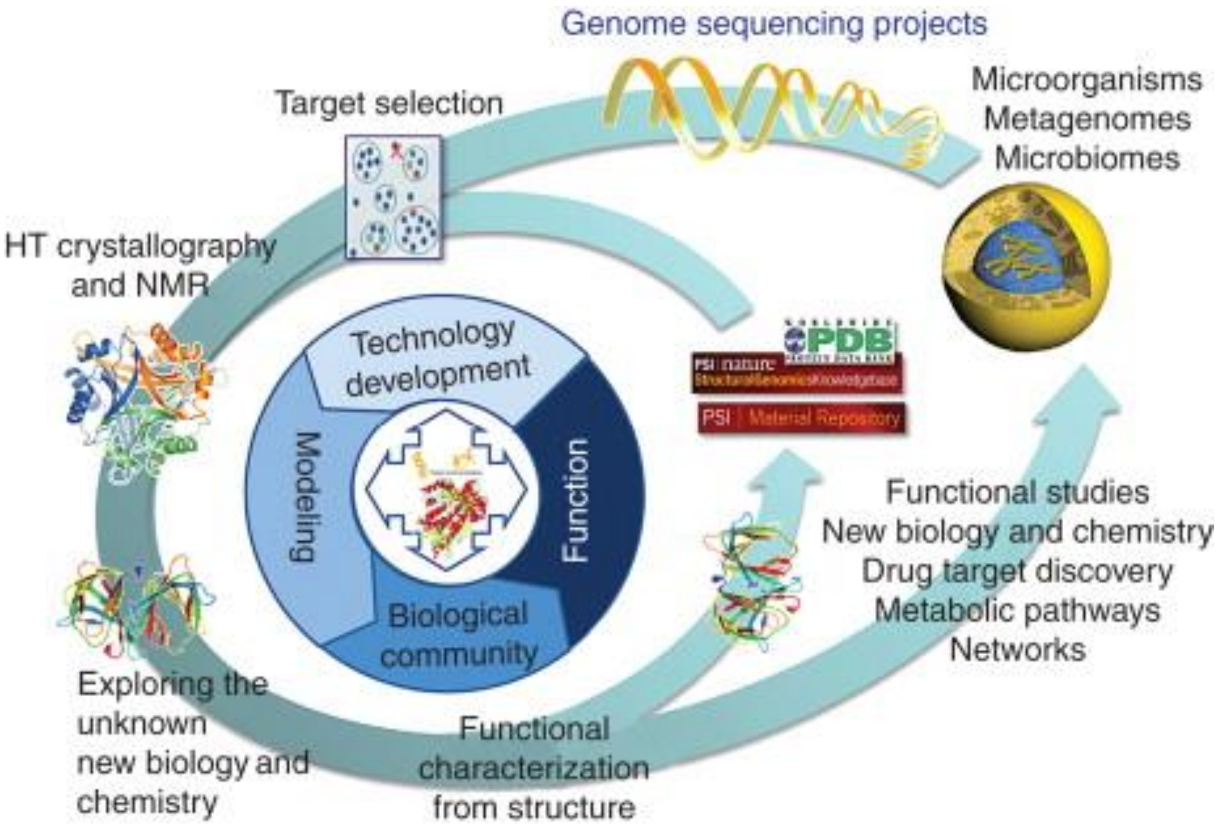
Challenges in solving protein structures



亿级

十万级



- **Growth of protein sequences and structure databases over time**

Figure from: https://www.dnastar.com

- **Genome Structure**
- **HT structural biology**

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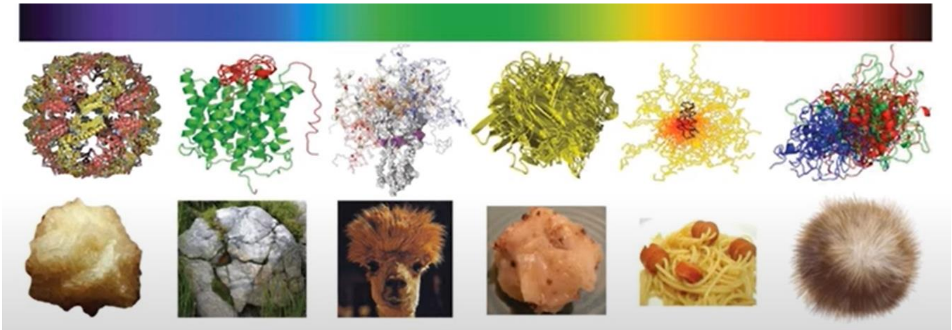
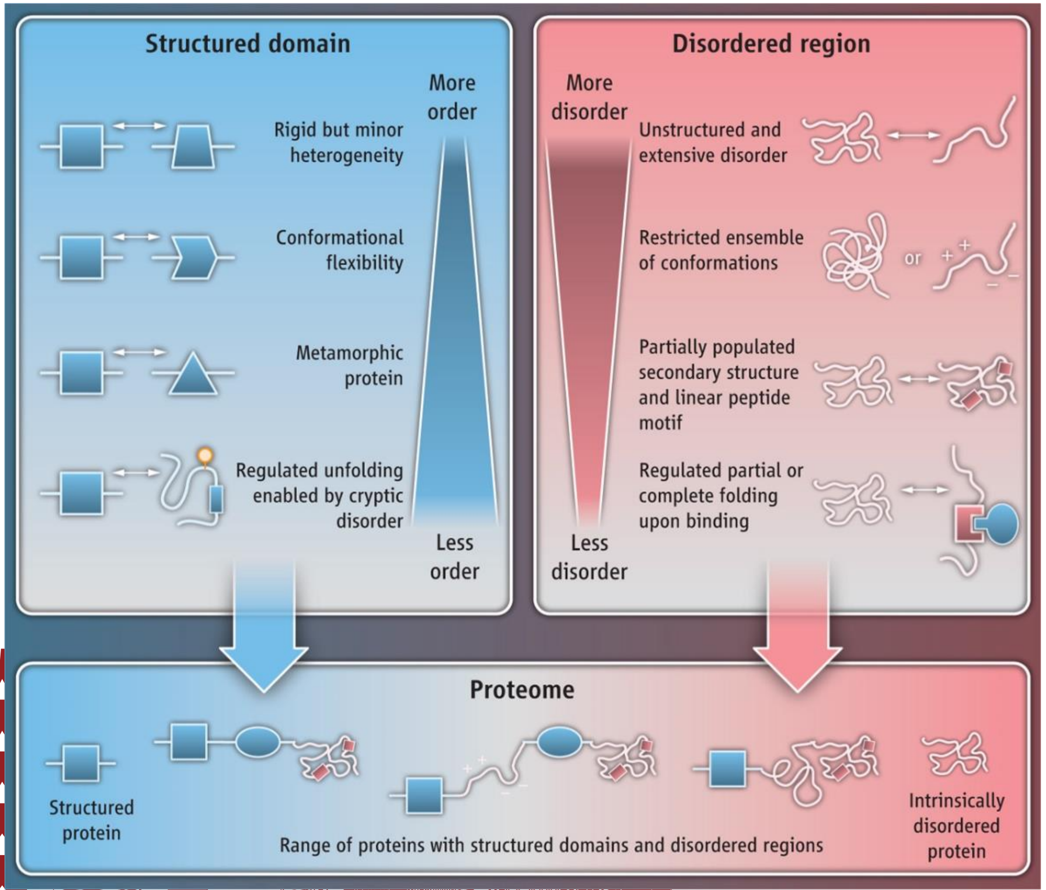
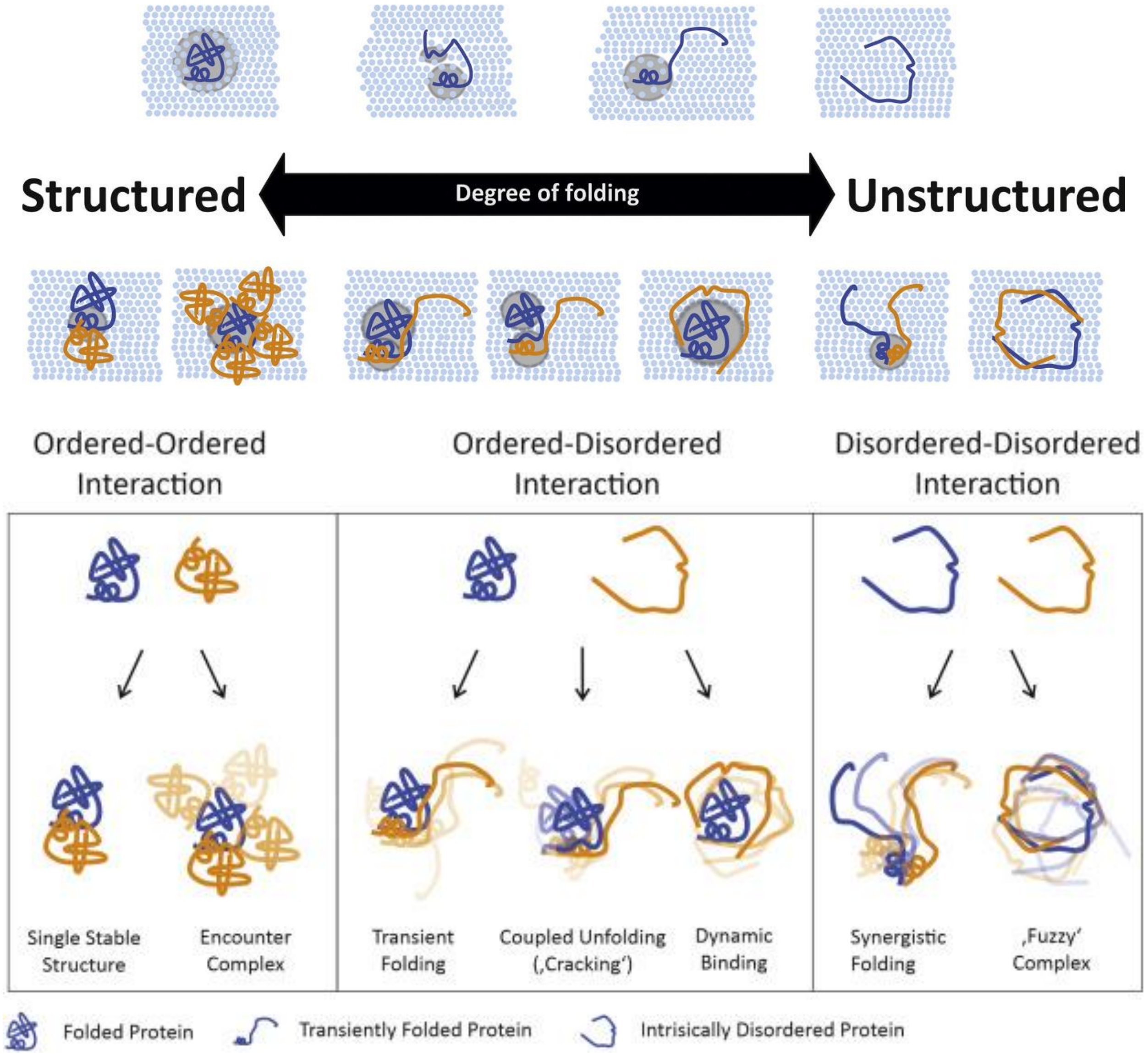
Challenges in disordered regions and proteins



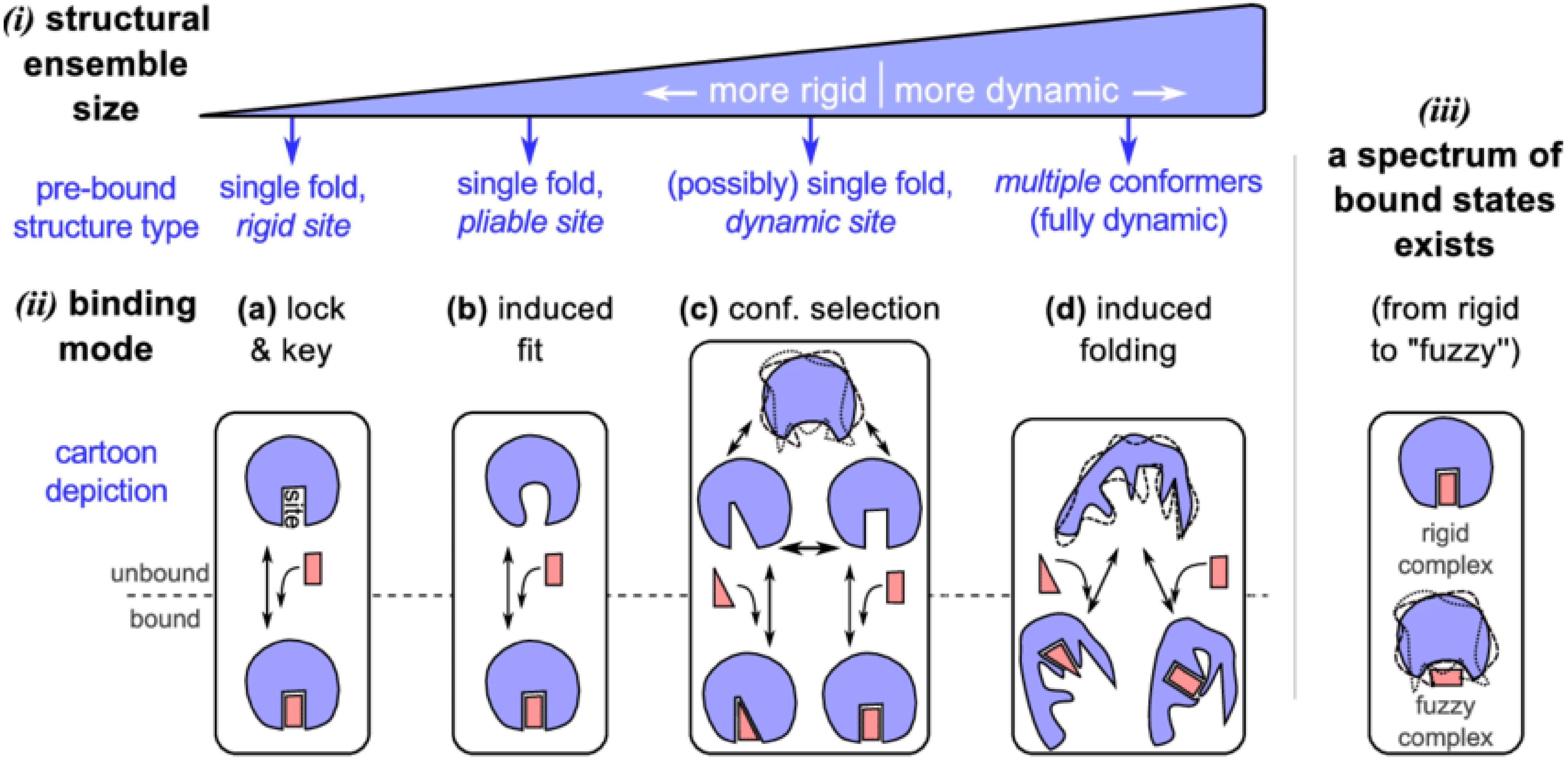




Babu et al. Science 2012;337:1460-1461

Figure (left top) from Shelly Deforte

The dynamic "new view" of protein structure and function



Proteomes, 2014, 2(1):128-153

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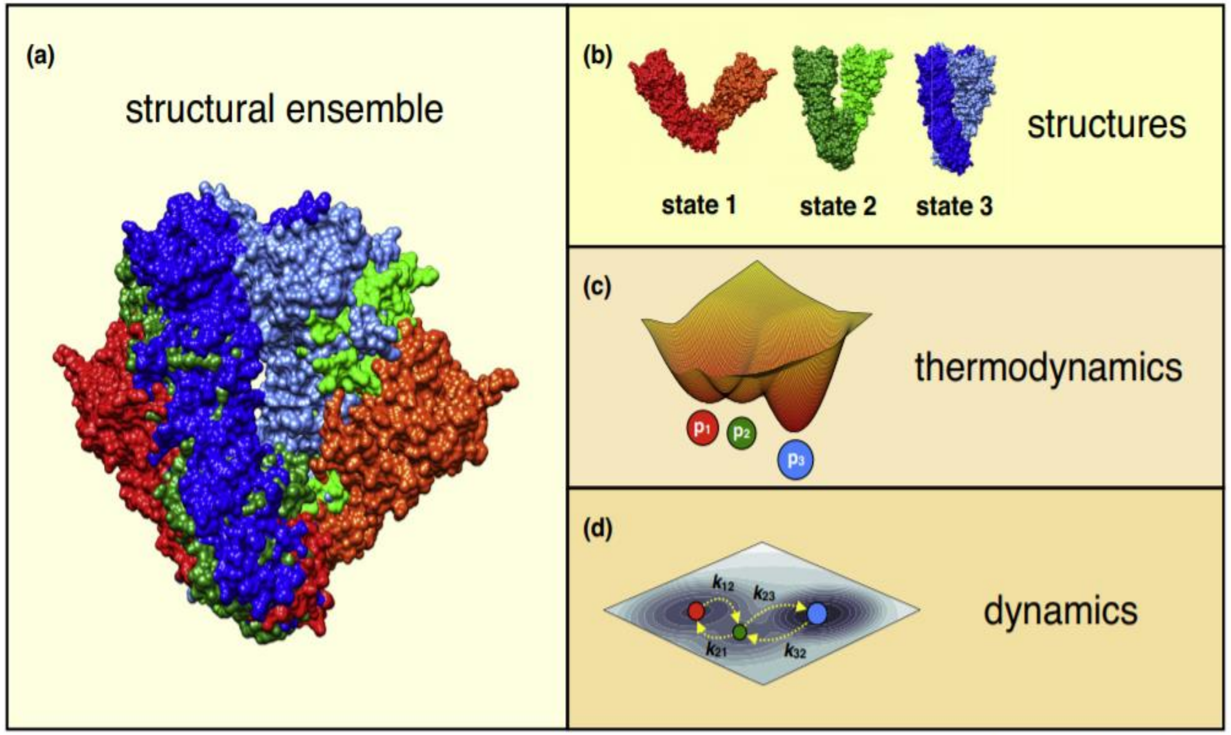
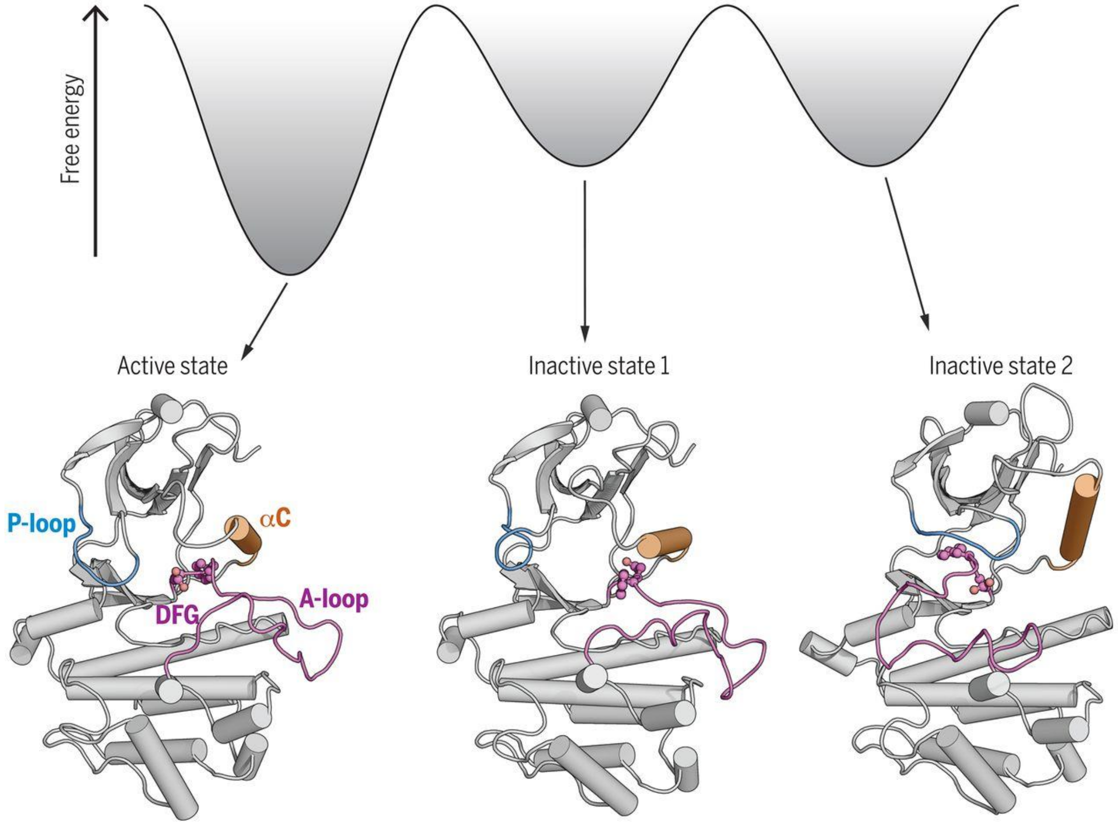
Why structural dynamic matters?

Protein structural ensembles provide unique insights to understand protein "dynamic" behavior



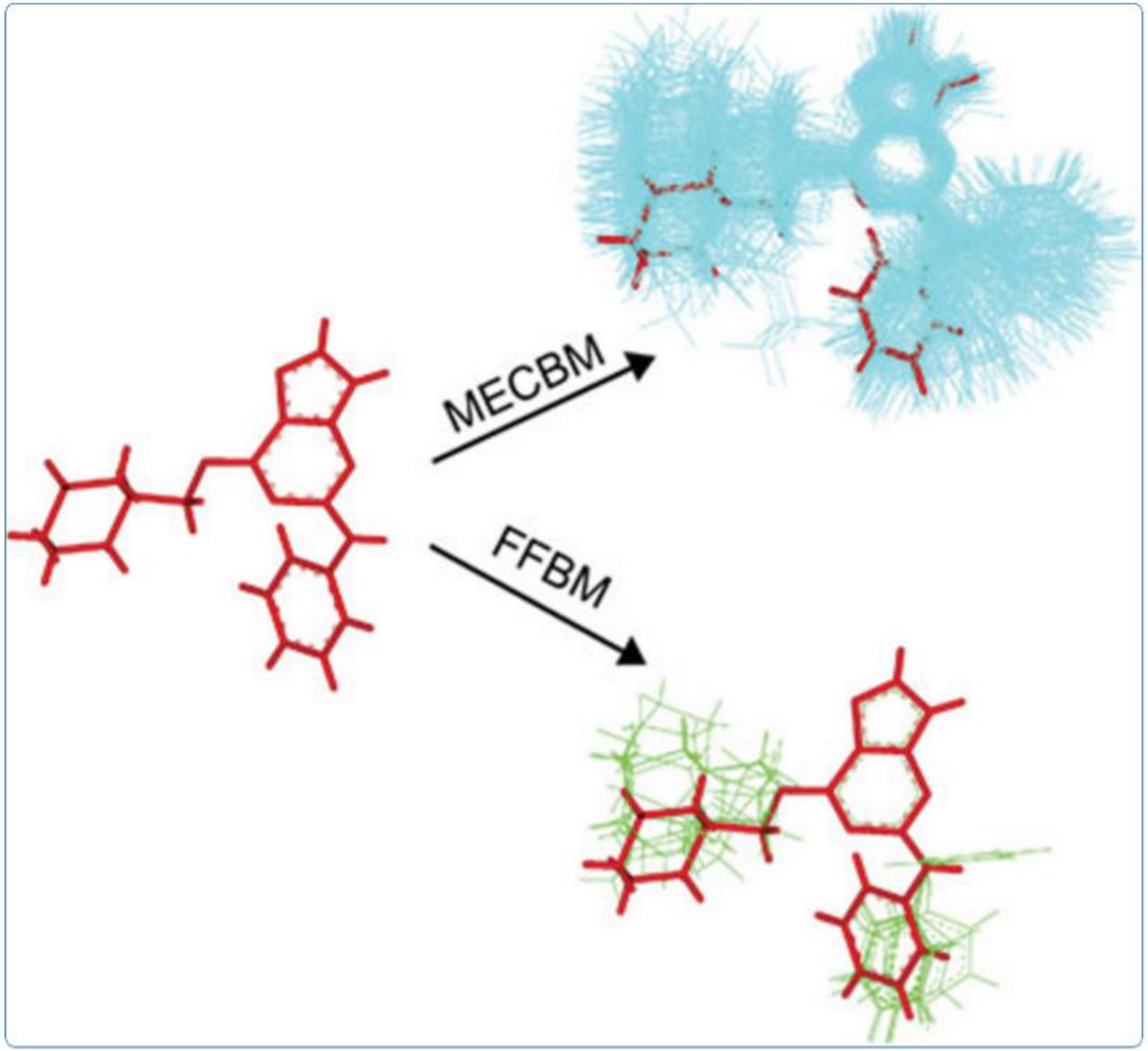
Current Opinion in Structural Biology

E.g. Transitions of Abl kinase between conformational states



Science, 2020, 370(6513):eabc2754

Small molecular conformational sampling



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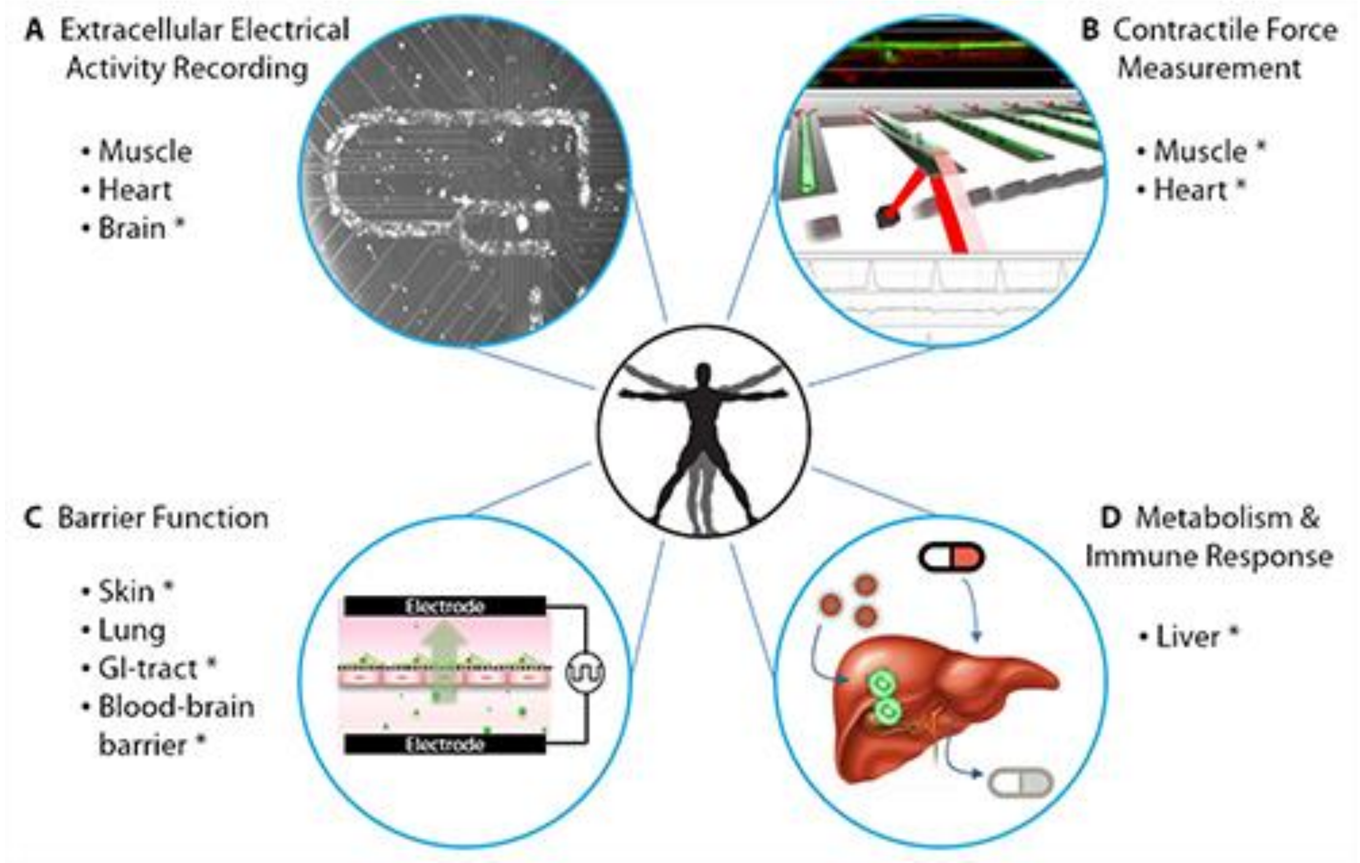
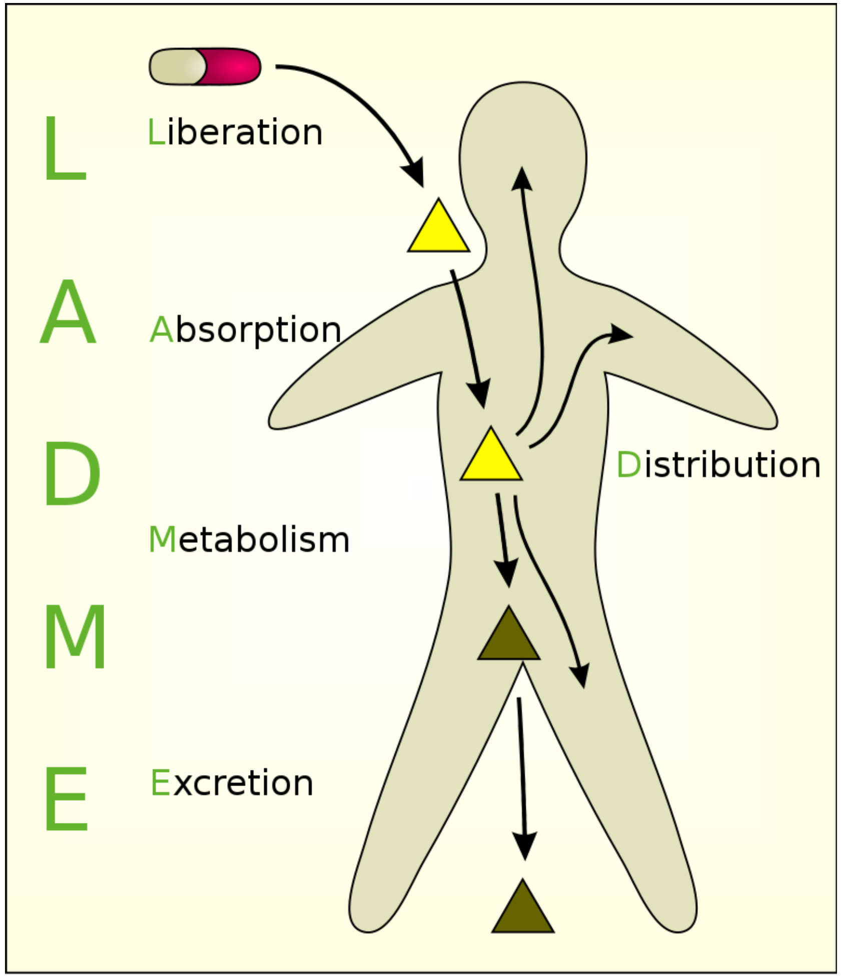
ADMET

ADMET: Turning chemicals into drugs

Optimization of Pharmacokinetics



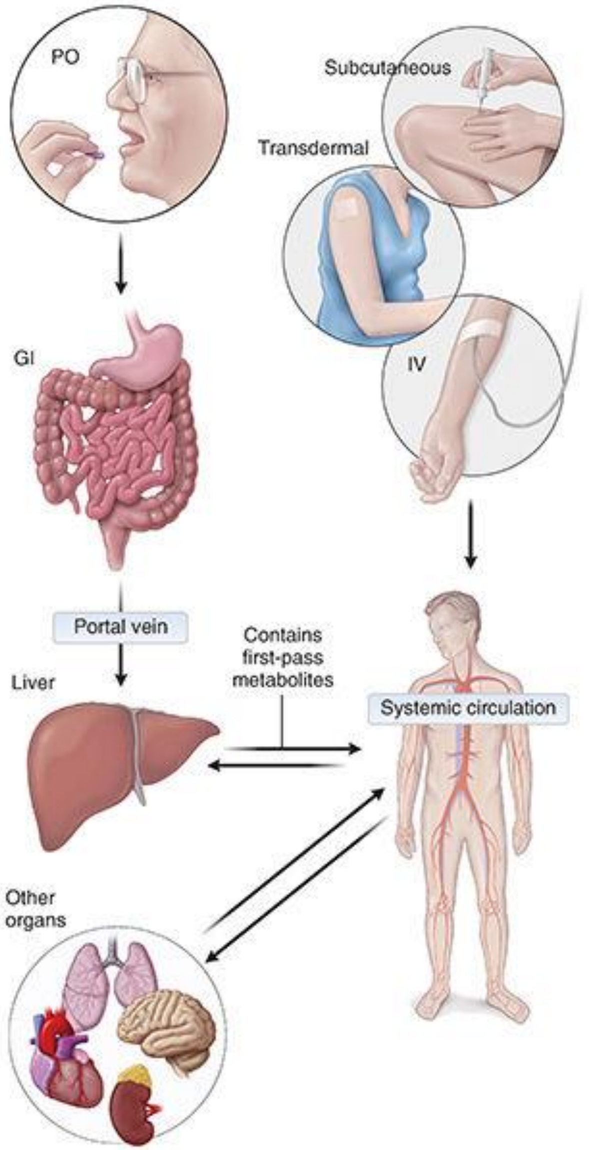


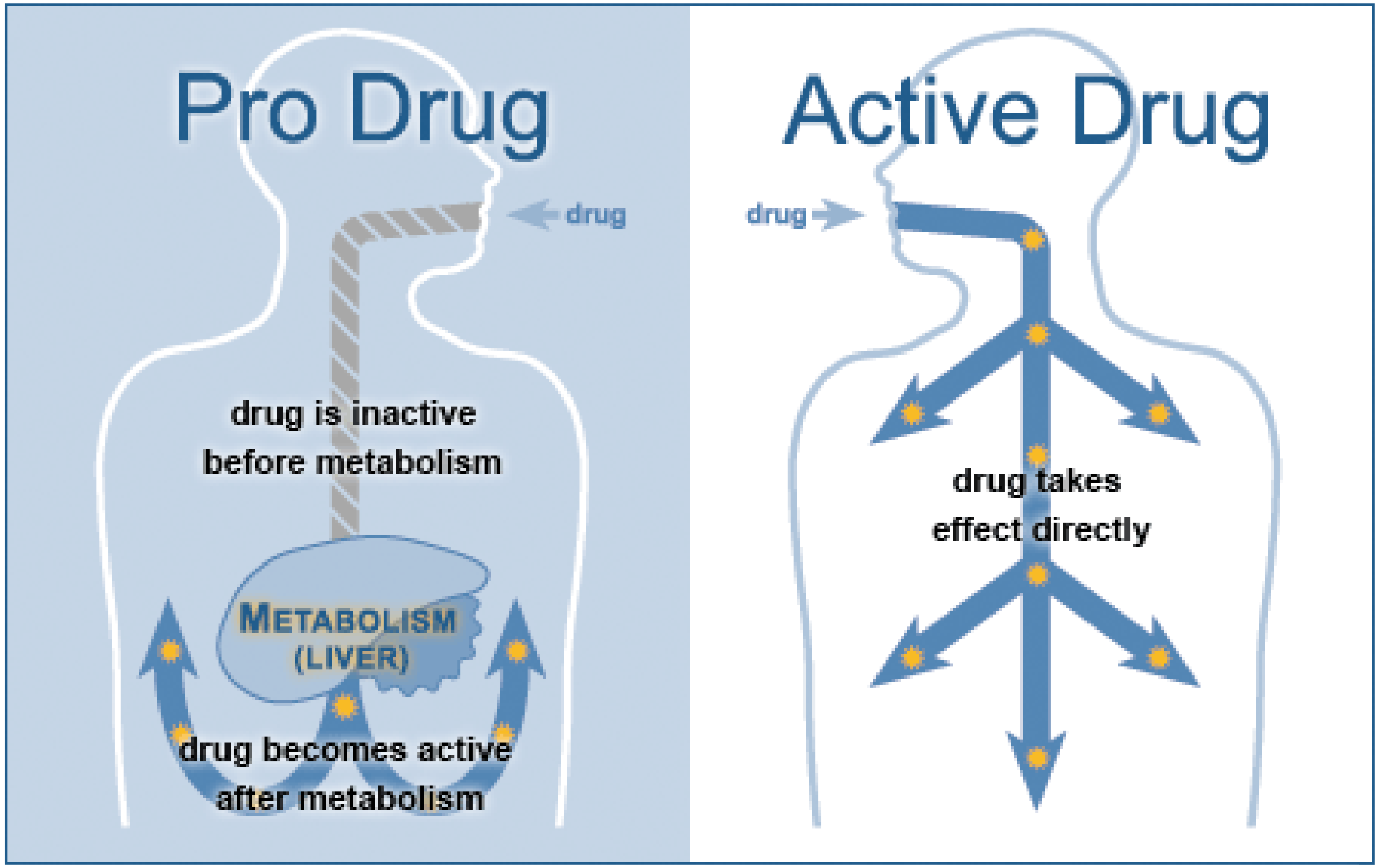
Liberation（释放）Absorption(吸收) Distribution(分布)

Metabolism（代谢） Excretion（排泄）Toxicity(毒性)

Toxicity

Drug metabolism





Biochemical or chemical process(es)

Prodrug *inactive* ⟶ ⟶ Drug *active*

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The Metabolic Rainbow: Deep Learning Phase I Metabolism in Five Colors

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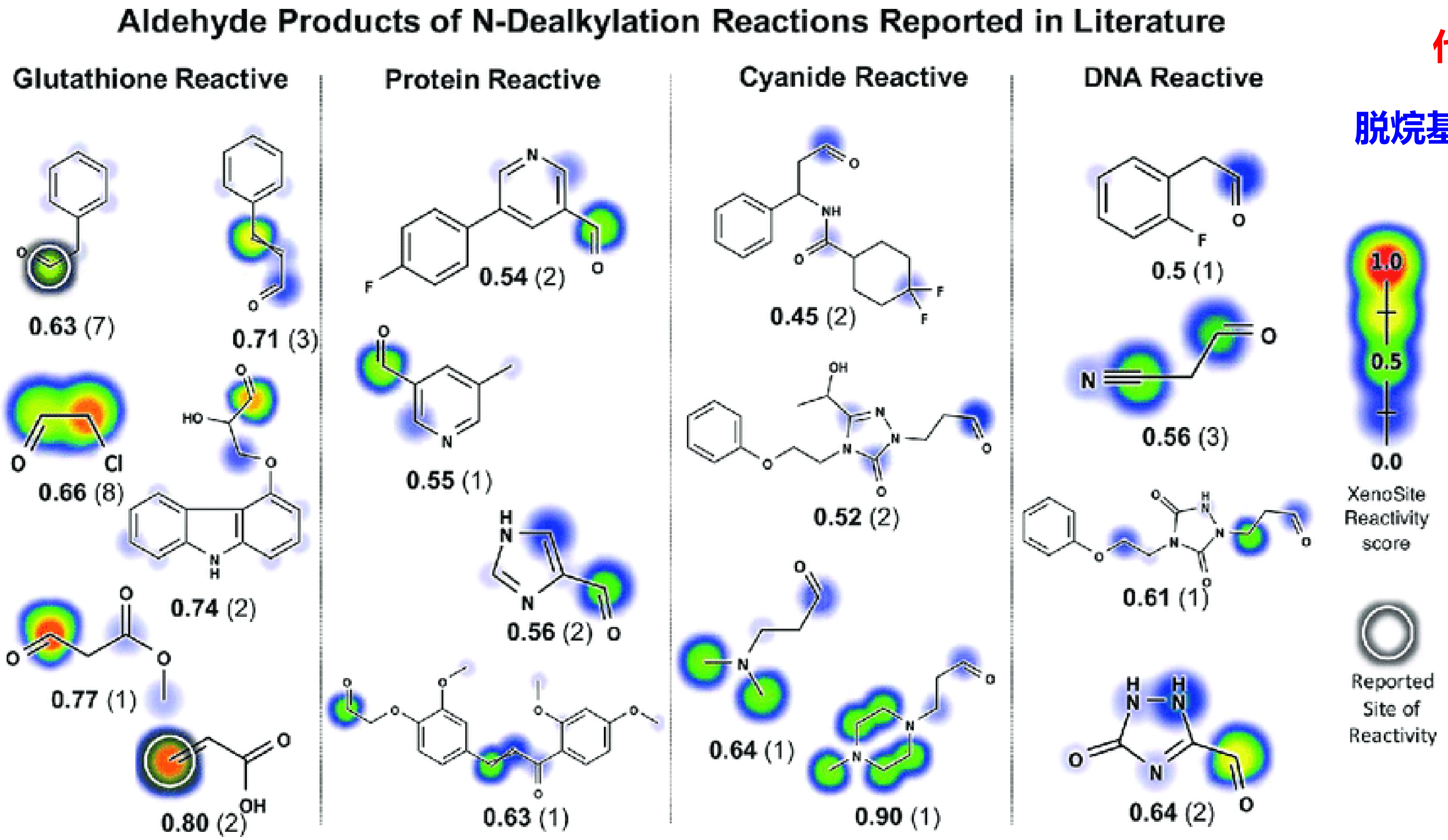
代谢彩虹

脱烷基化的醛产物



J. Chem. Inf. Model. 2020, 60, 3, 1146–1164

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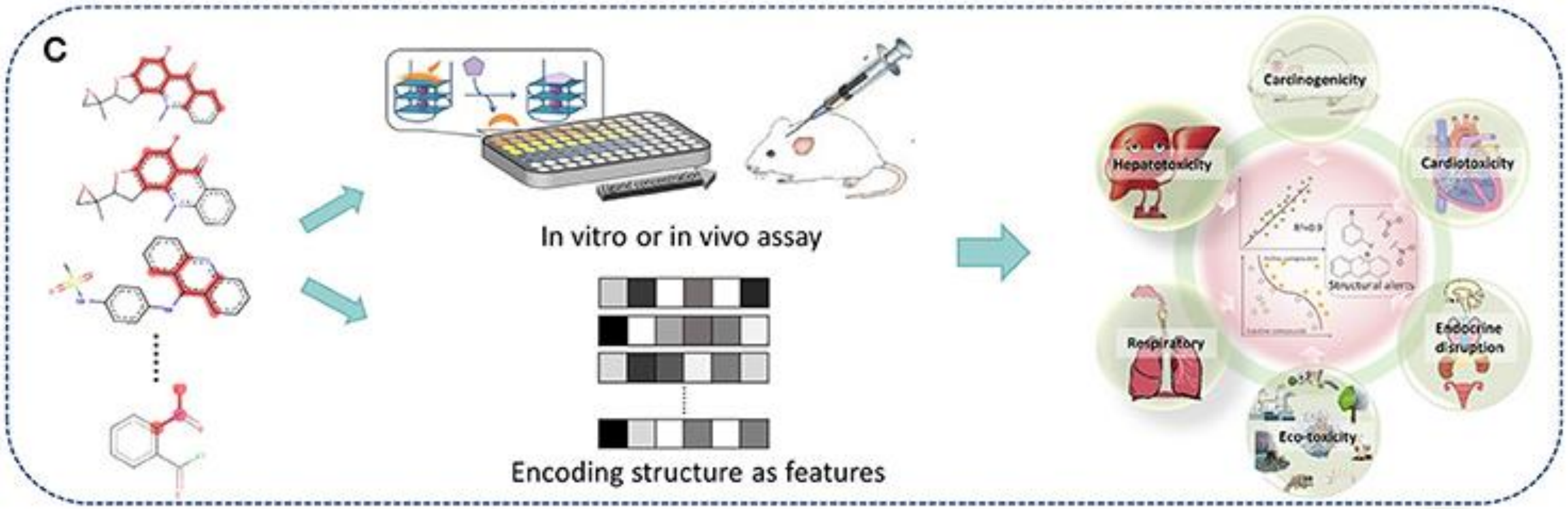
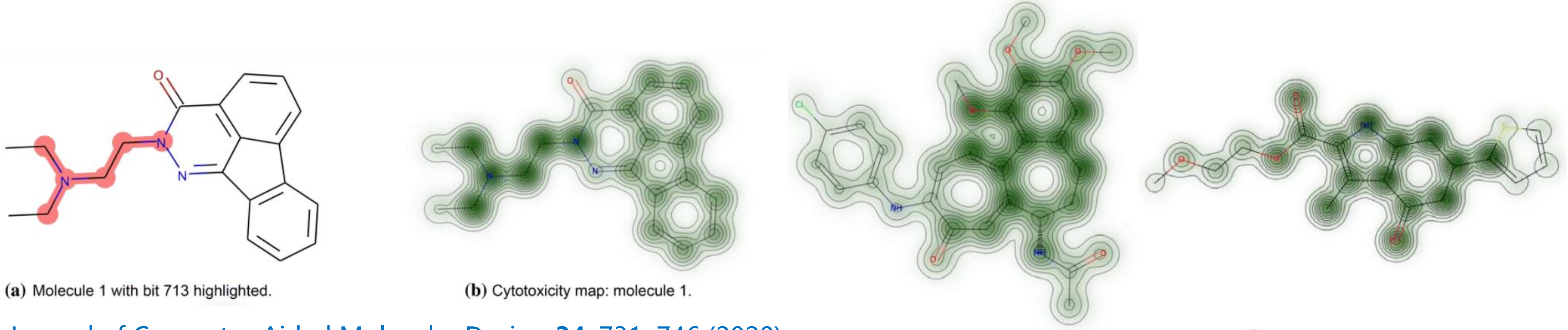
Revealing cytotoxic substructures in molecules using deep learning



Front. Chem. 2018, 6:30



(a) Molecule 1 with bit 713 highlighted.

(b) Cytotoxicity map: molecule 1.

(c) Cytotoxicity map: molecule 2B.

(d) Cytotoxicity map: molecule 3A.

Journal of Computer-Aided Molecular Design **34**, 731–746 (2020)

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End of Lecture 21